

Duration: 120 min

College of Medicine

Grades: 1

Lect no. 1

Introduction to human anatomy

Hawler Medical University
College of Medicine
Department of Anatomy and Pathology



Head lines

- Definitions of anatomy.
- Descriptive Anatomical position.
- Terms related to position.
- Terms related to *Movements*.
- Skin
- Fascia

By the end of lecture you should know

- ✓ **What's anatomy and its branches?**
- ✓ **Main imaginary planes**
- ✓ **What's anatomical position**
- ✓ **Differences between regional and systemic anatomy**
- ✓ **Anatomical terms**
- ✓ **Body cavities**
- ✓ **Skin**
- ✓ **Fascia**

Anatomy
vs
Dissection

*Clayton
April 20 1877
Winchester*

Anatomy

- Is the **oldest** scientific discipline of medicine.
- Is the scientific study of the **structure** and **Function** of the body

Ana= up

Tomy= cutting

Discovering

Anatomy

- An anatomist is concerned with the **shape, size, position, structure, blood supply and innervation.**

Anatomy

Includes those
structures that
can be seen
grossly



Branches of Anatomy

- 1 Gross (macroscopic) Anatomy**
 - a. Regional Gross Anatomy
 - b. Systemic Gross Anatomy
- 2 Microscopic Anatomy:**
 - a. Histology
 - b. Cytology

Branches of Anatomy

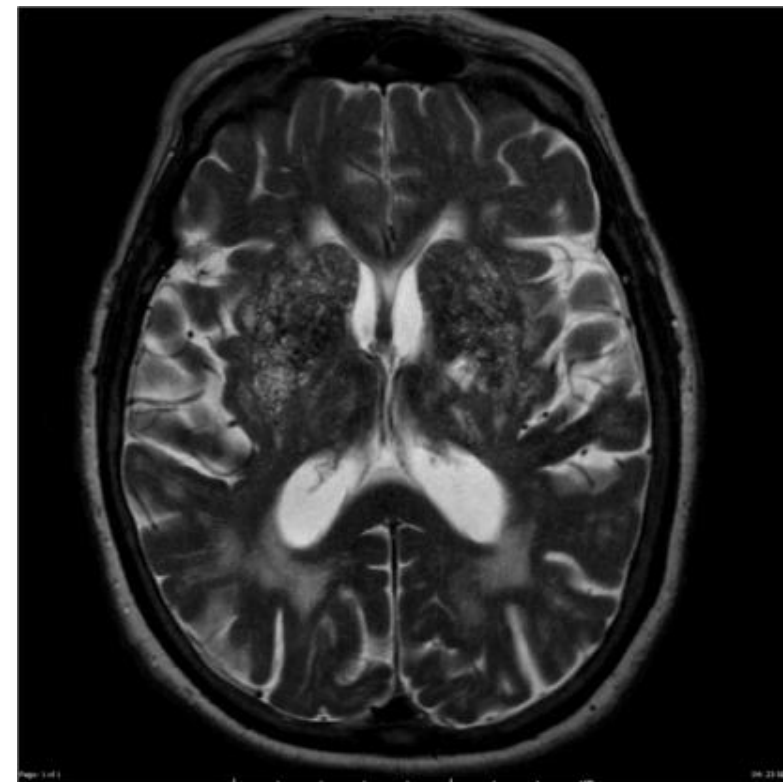
3 Surface anatomy:

- deals with **anatomical** features that can be studied by sight, without dissection.
- **surface anatomy of the chest** is important because it is one of the areas most frequently subjected to *physical examination*, like *auscultation* and *percussion*.

Branches of Anatomy

4 Radiological anatomy:

- which involves the study of anatomy through the use of radiographic films i.e the X-Ray.



5 Embryology

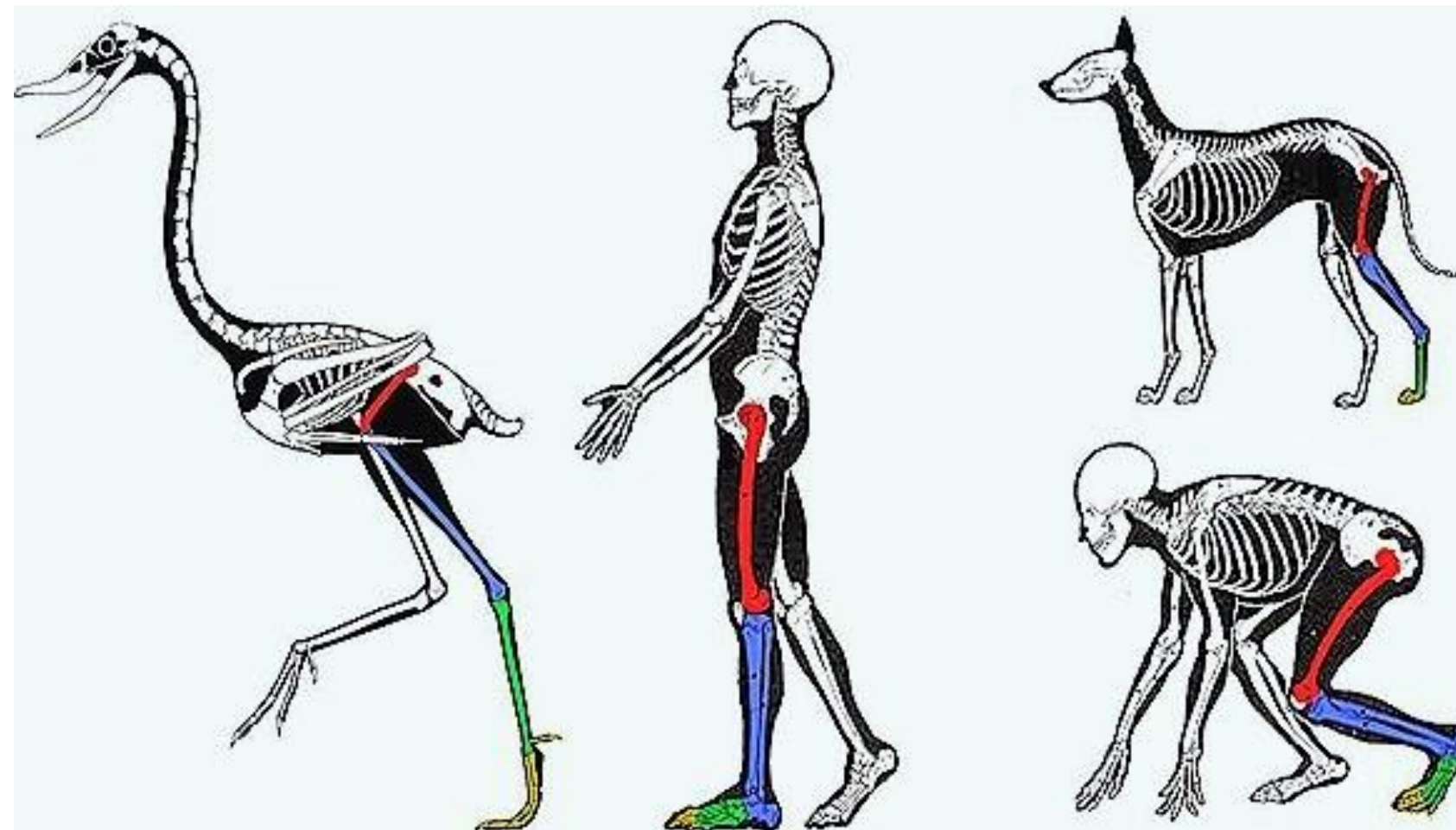
- Refers to the **development** and formation of the **parts of body** *from fertilization to delivery.*



Branches of Anatomy

6 Comparative anatomy:

- is the study of similarities and differences in the **anatomy** of different species.



Branches of Anatomy

7 Basic anatomy:

- is the study of the minimal amount of anatomy consistent with the **understanding of the overall structure and function of the body.**

8 Clinical anatomy:

- is the study of the macroscopic structure and function of the body as it relates to the practice of medicine and other health sciences.



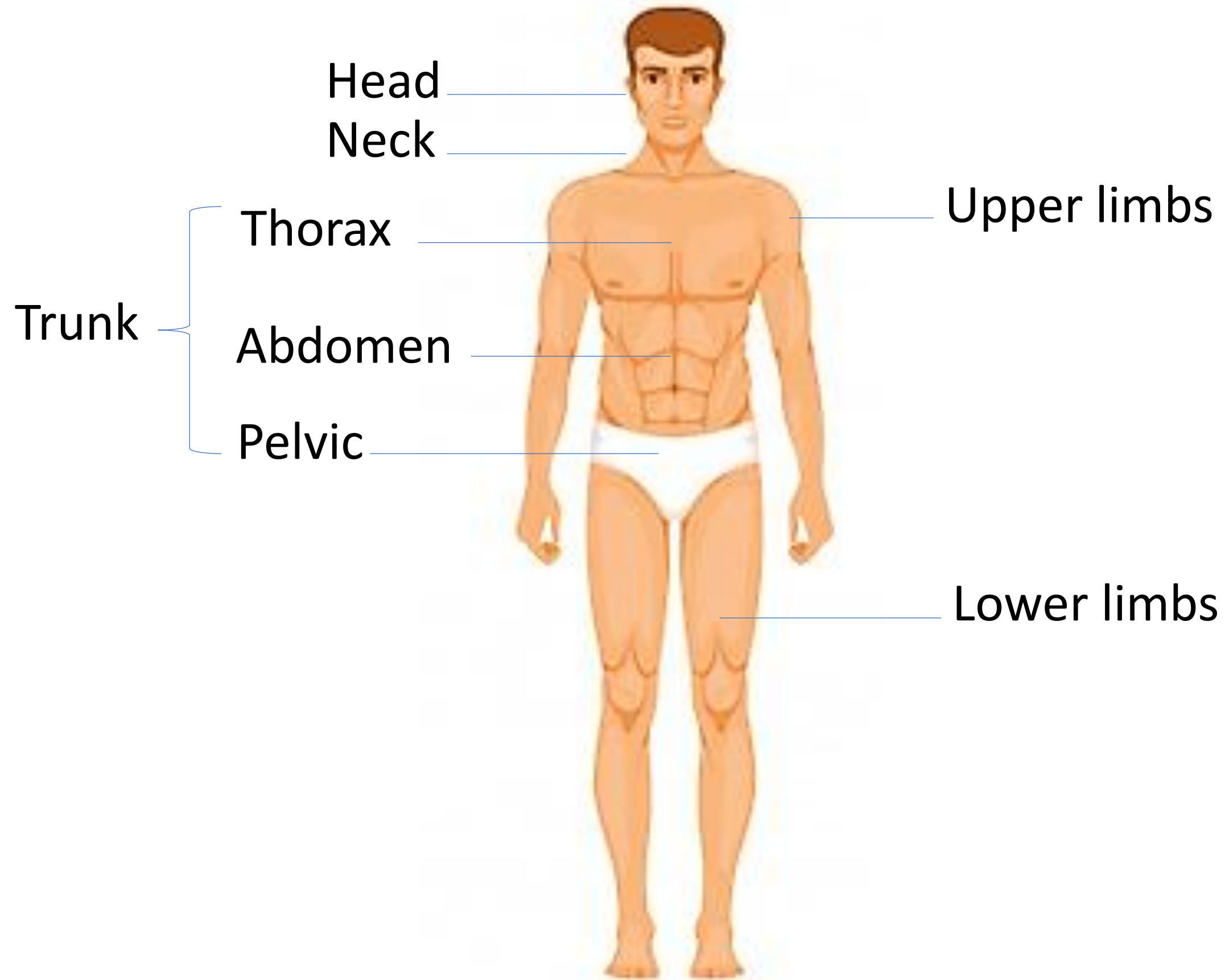
Study of Anatomy

- There are **two main ways** to studying anatomy:

- ① by **Region approach**.

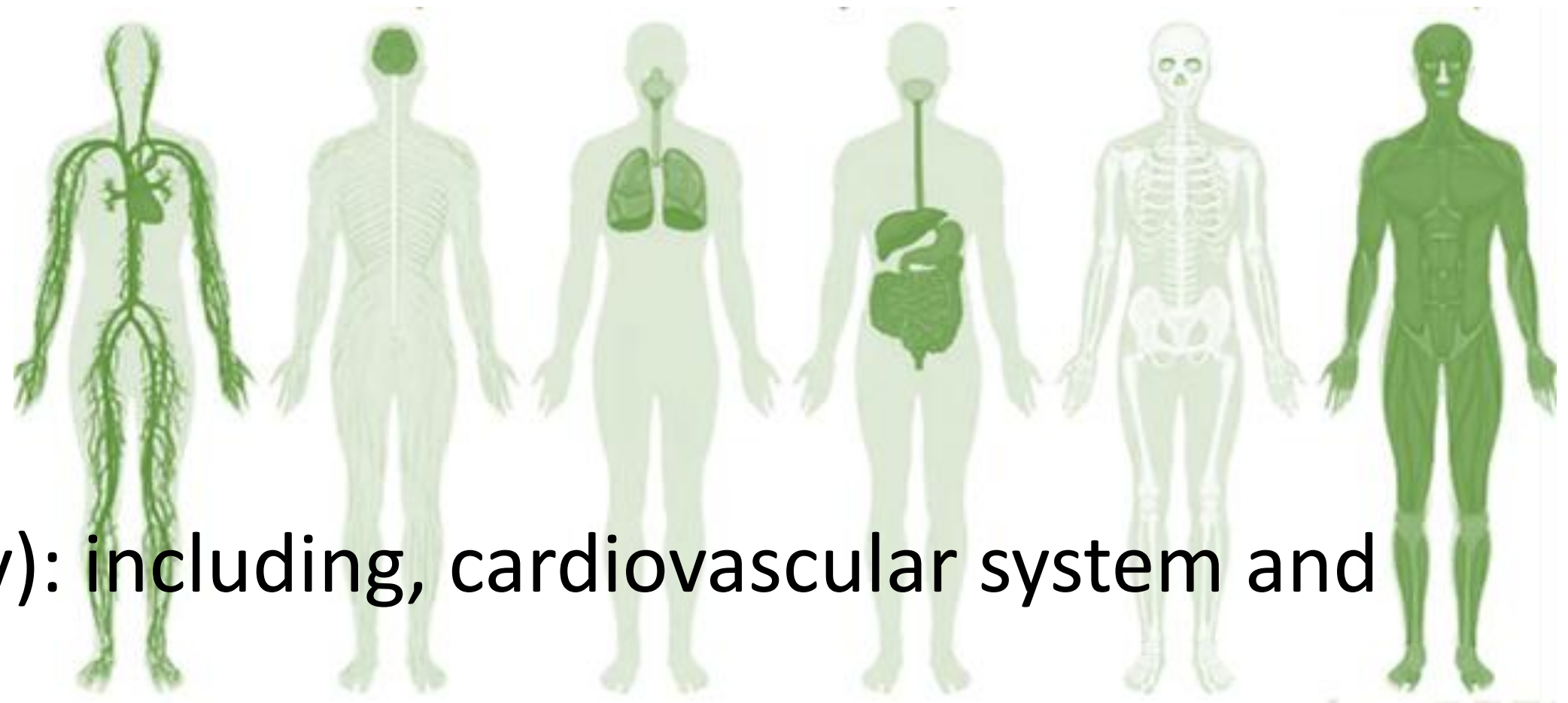
- ② by **System approach**.

Regional Anatomy



Systemic Anatomy

1. Integumentary system (dermatology).
2. Skeletal system.(osteology).
3. Articular system.(arthrology).
4. Muscular system (myology).
5. Nervous system (neurology).
6. Circulatory system (Angiology): including, cardiovascular system and lymphatic system.
7. Digestive or alimentary system (gastroenterology)



Systemic Anatomy

8. Respiratory system (pulmonology).
9. Urinary system (urology).
10. Reproductive or genital system: Including, gynecology for females and andrology for males.
11. Endocrine system (endocrinology).

Anatomical position

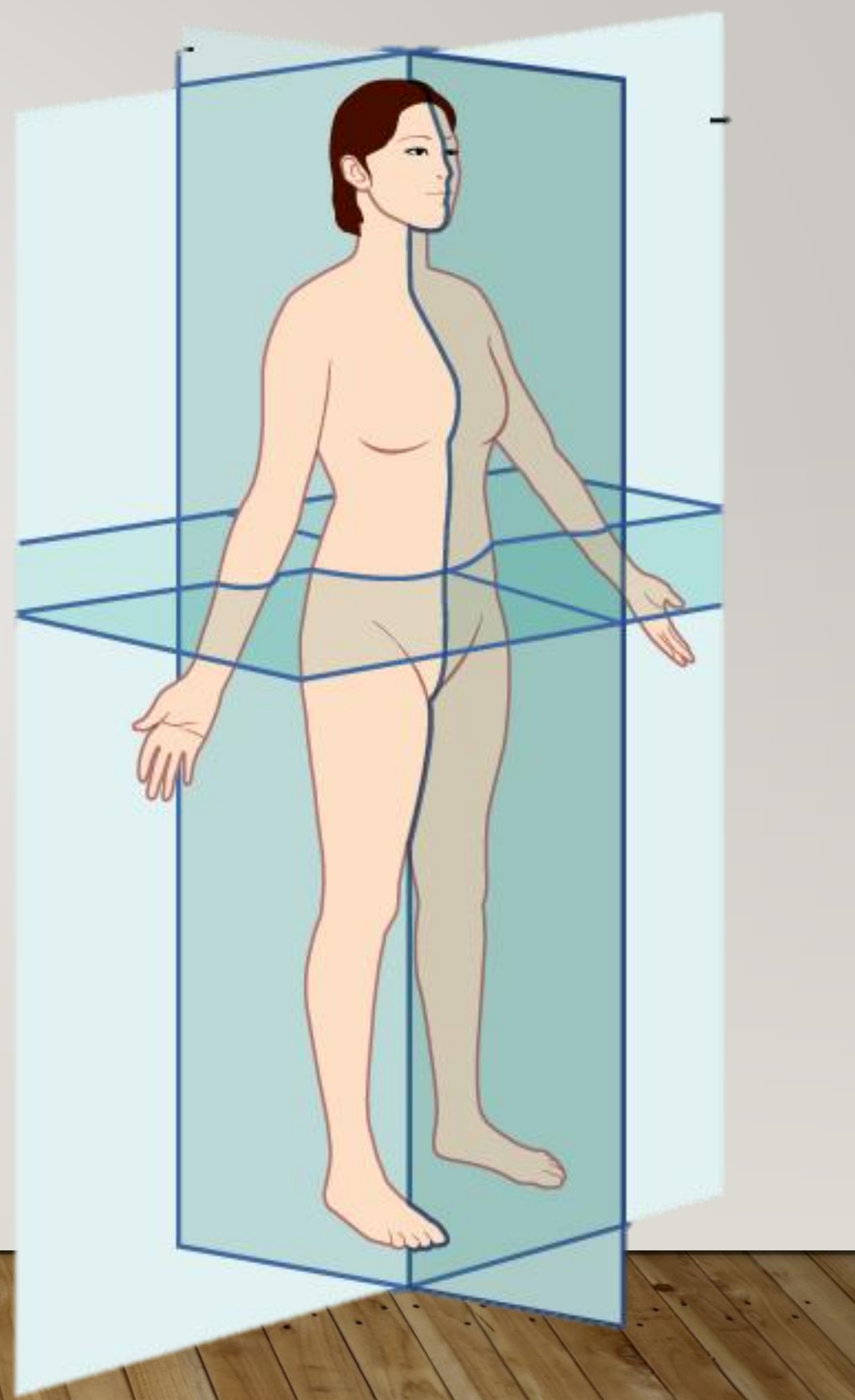


Anatomical position

All description of the human body are based on the assumption that the person is:

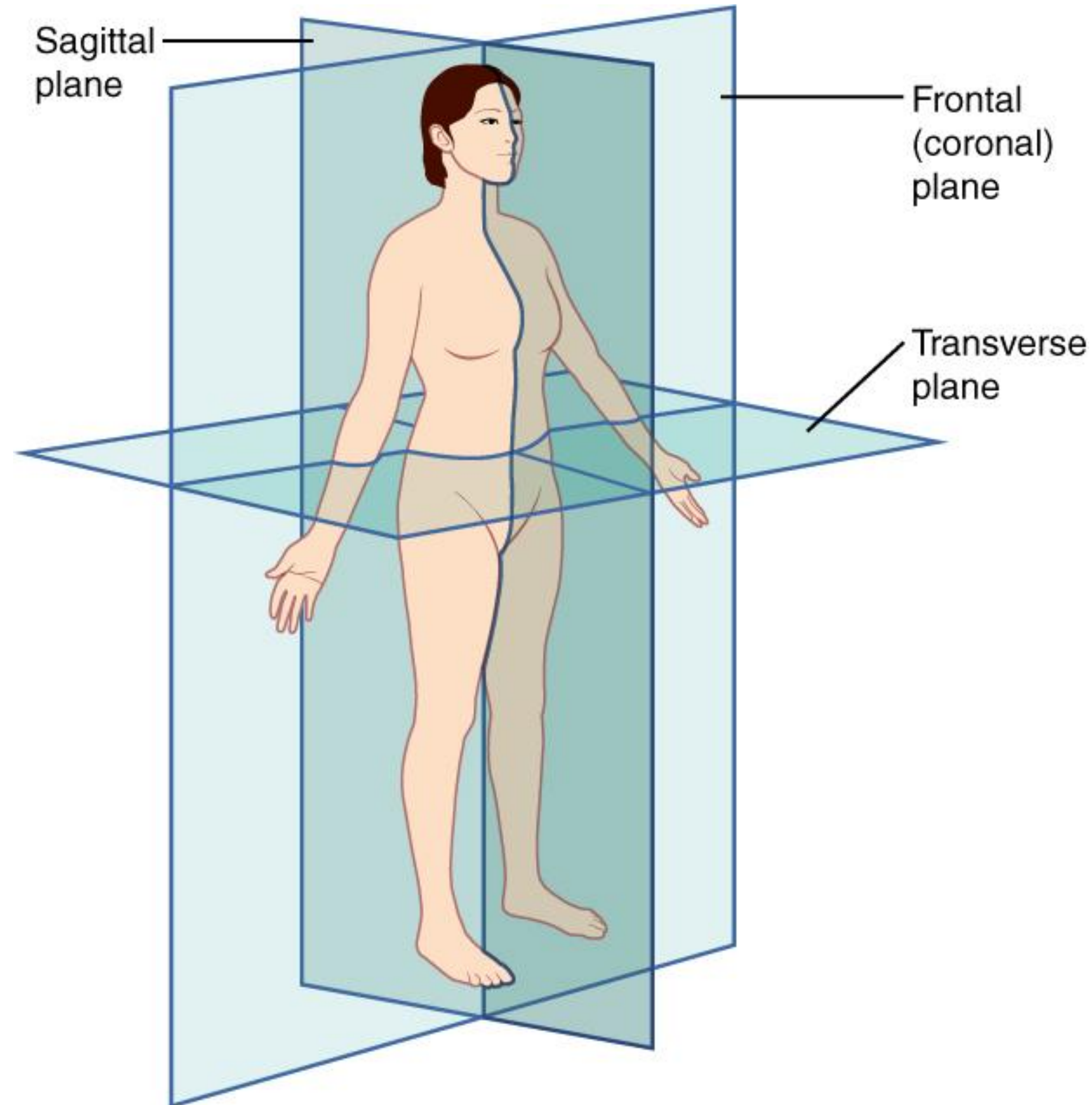
- i. Standing erect position.
- ii. Upper limbs by the sides.
- iii. Face and palms of the hands directed forward.

Descriptive Anatomical Terms

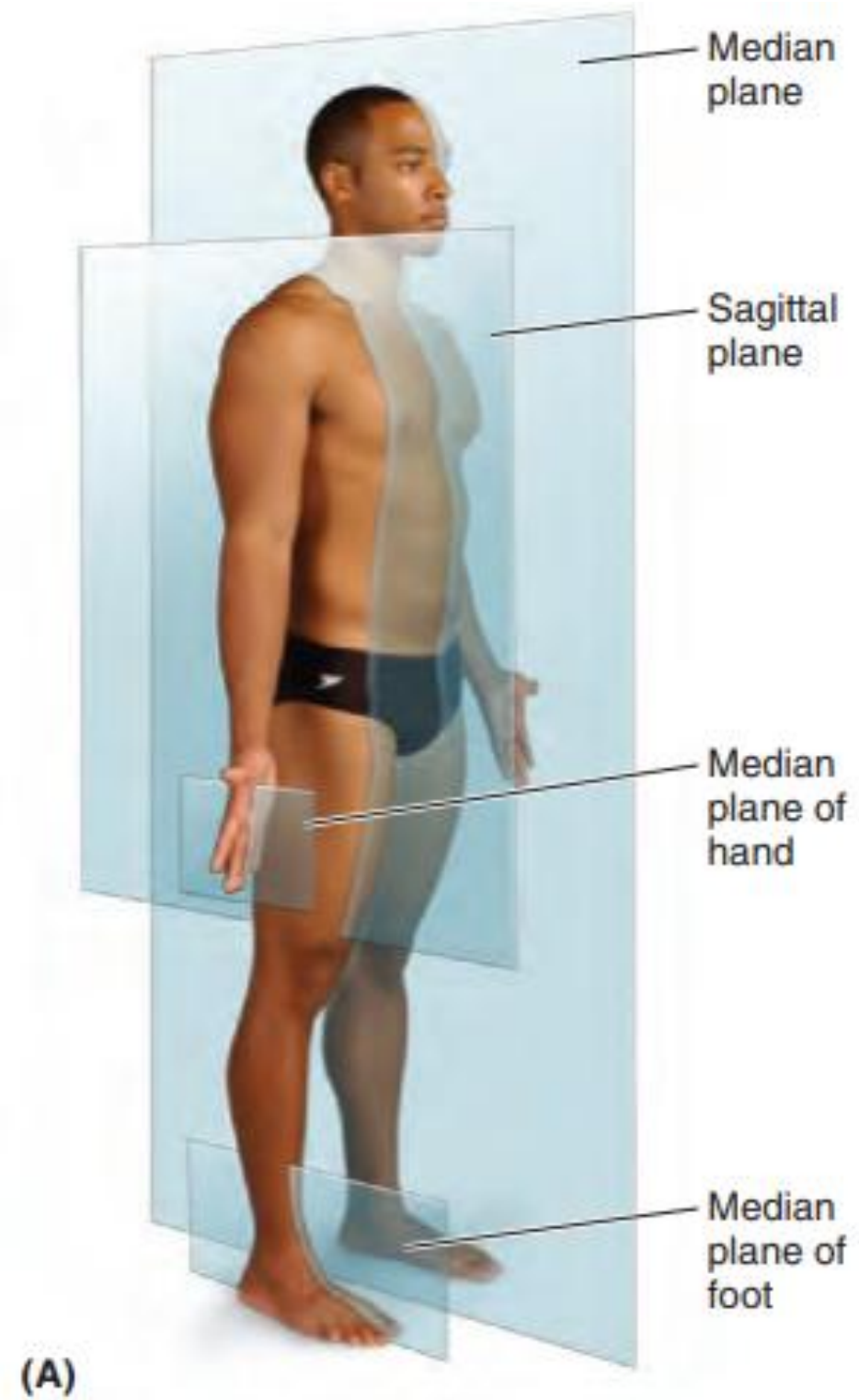


Body planes

- 1 Median plane
- 2 Sagittal plane.
- 3 Coronal plane.
- 4 Transverse plane.

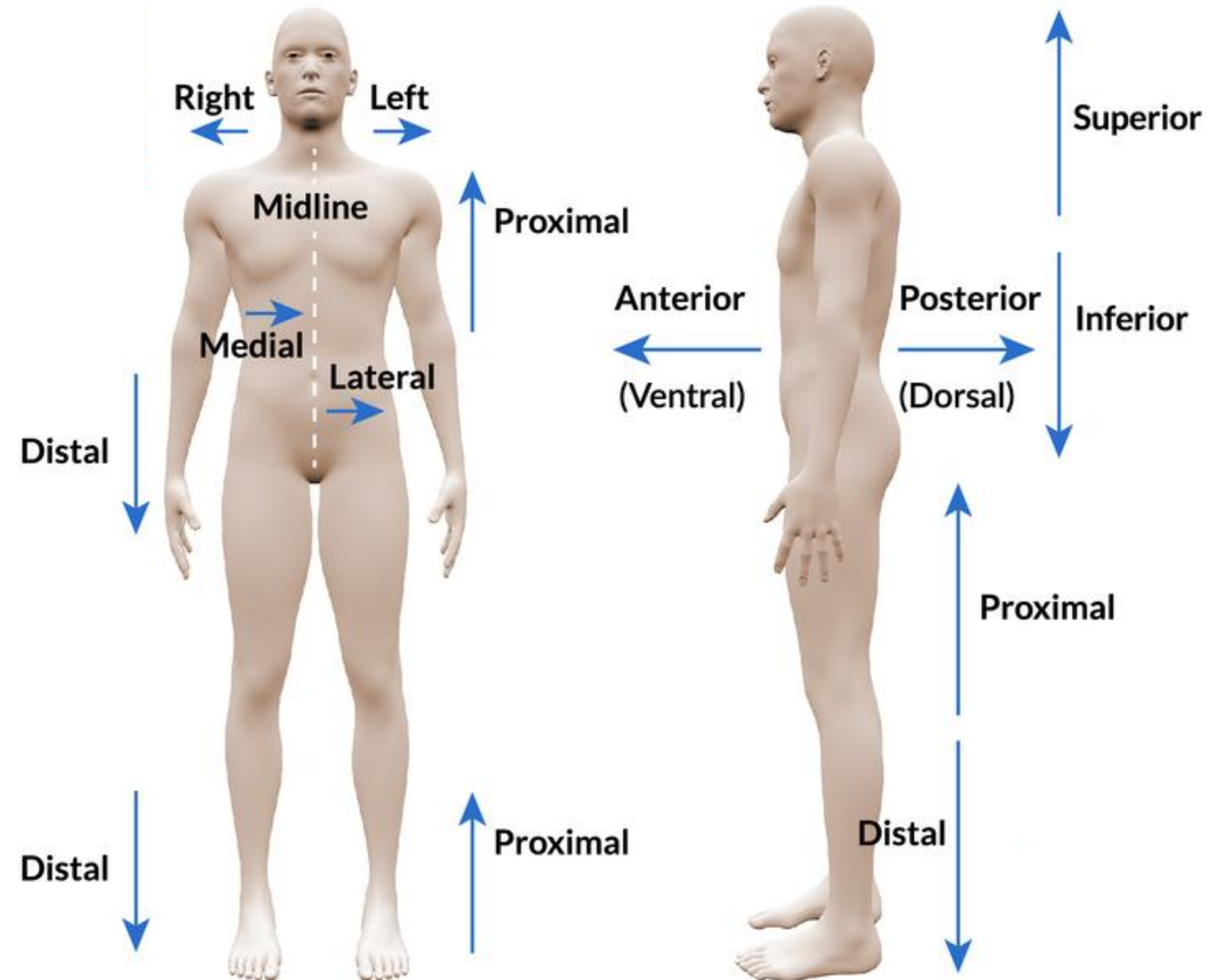


Body planes



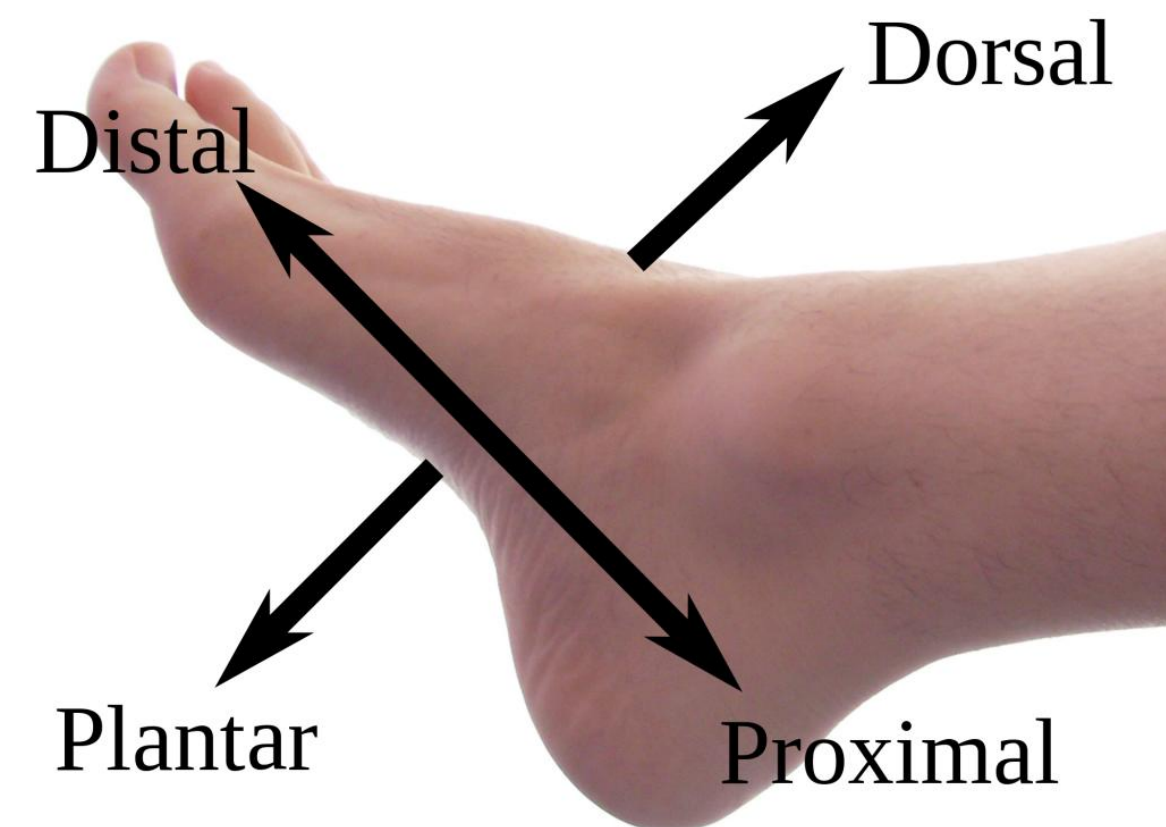
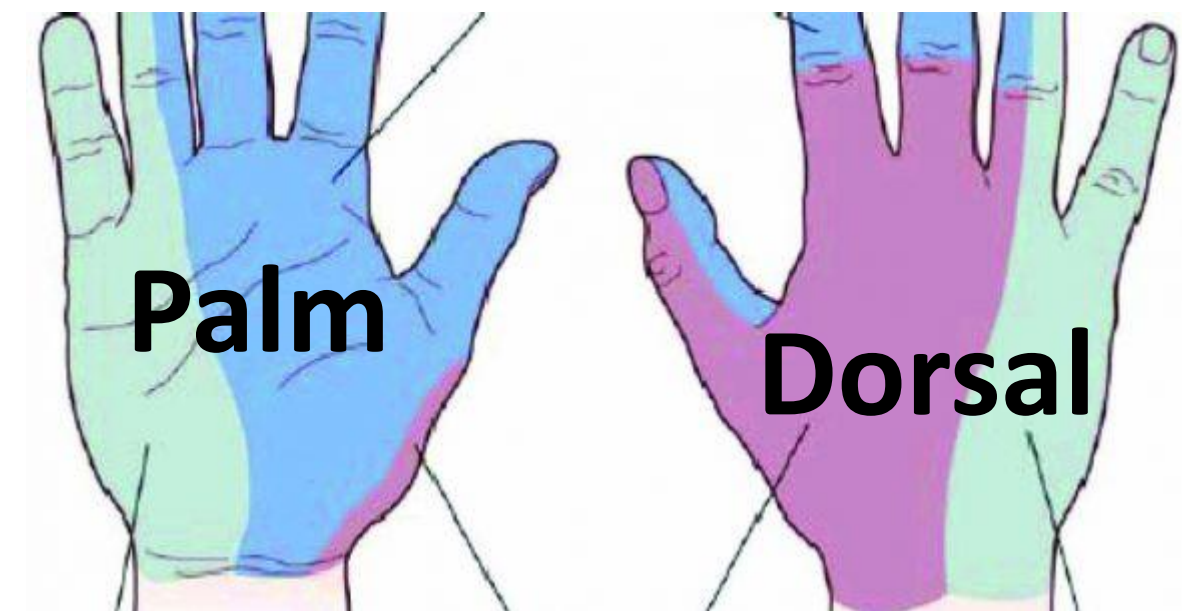
Terms of position (Directional Terms)

- **ANTERIOR** :
Front of the body.
- **POSTERIOR** :
Back of the body.



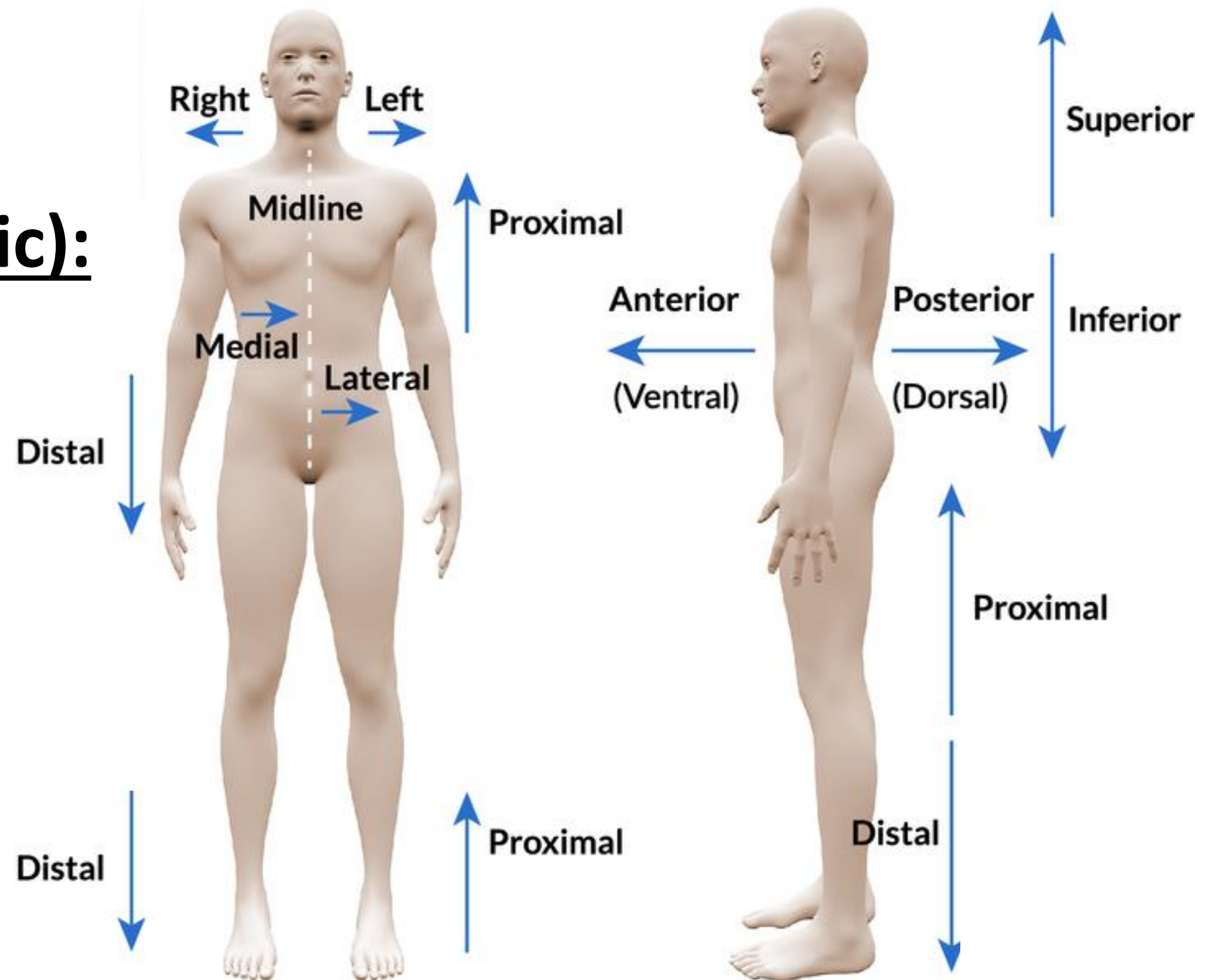
Terms of position (Directional Terms)

In the foot, **dorsal**
and **plantar** are
more specific
than superior and
inferior.



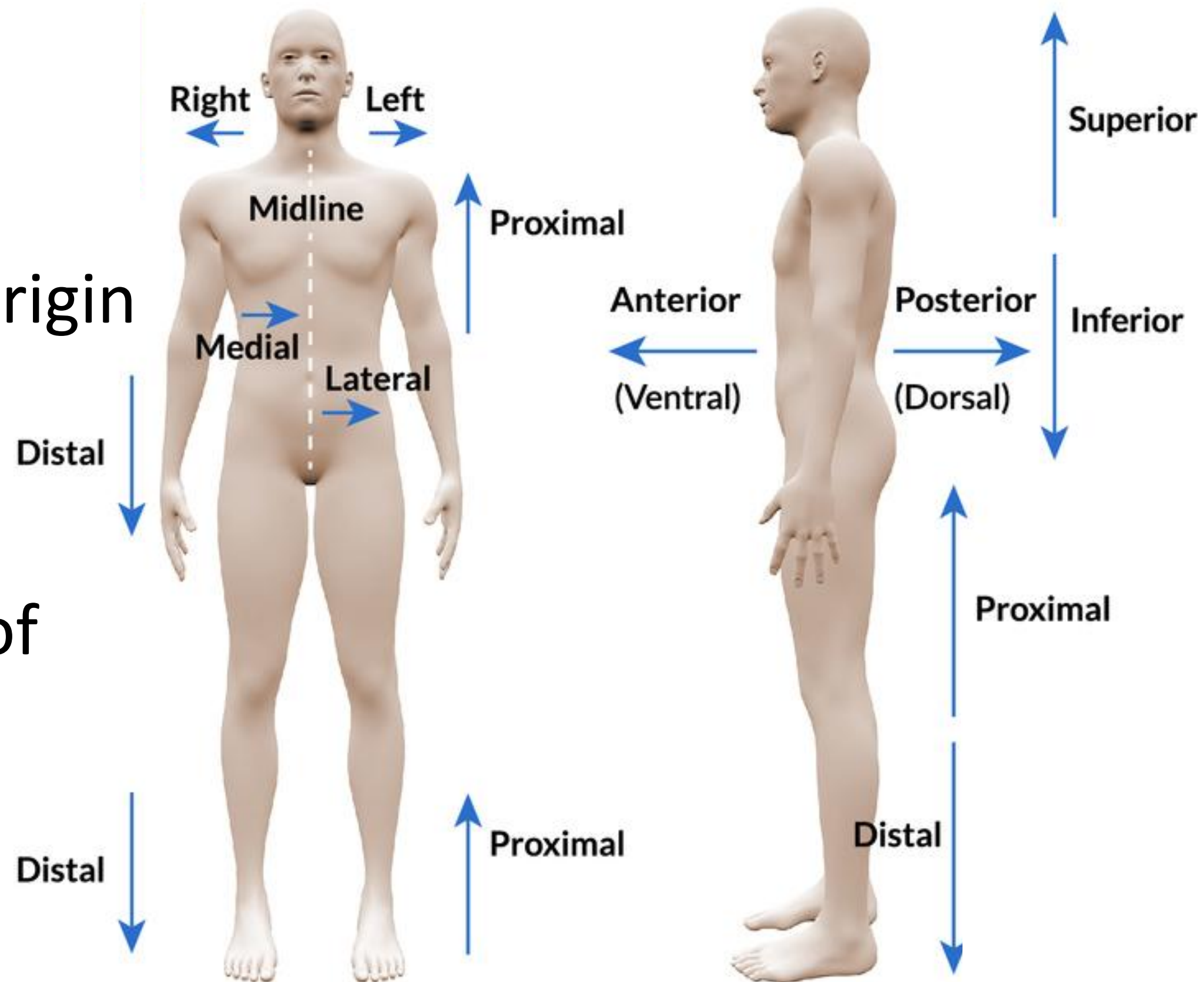
Terms of position (Directional Terms)

- **SUPERIOR (Cranial/Cephalic):**
toward the head
- **INFERIOR (Caudal):**
away from the head



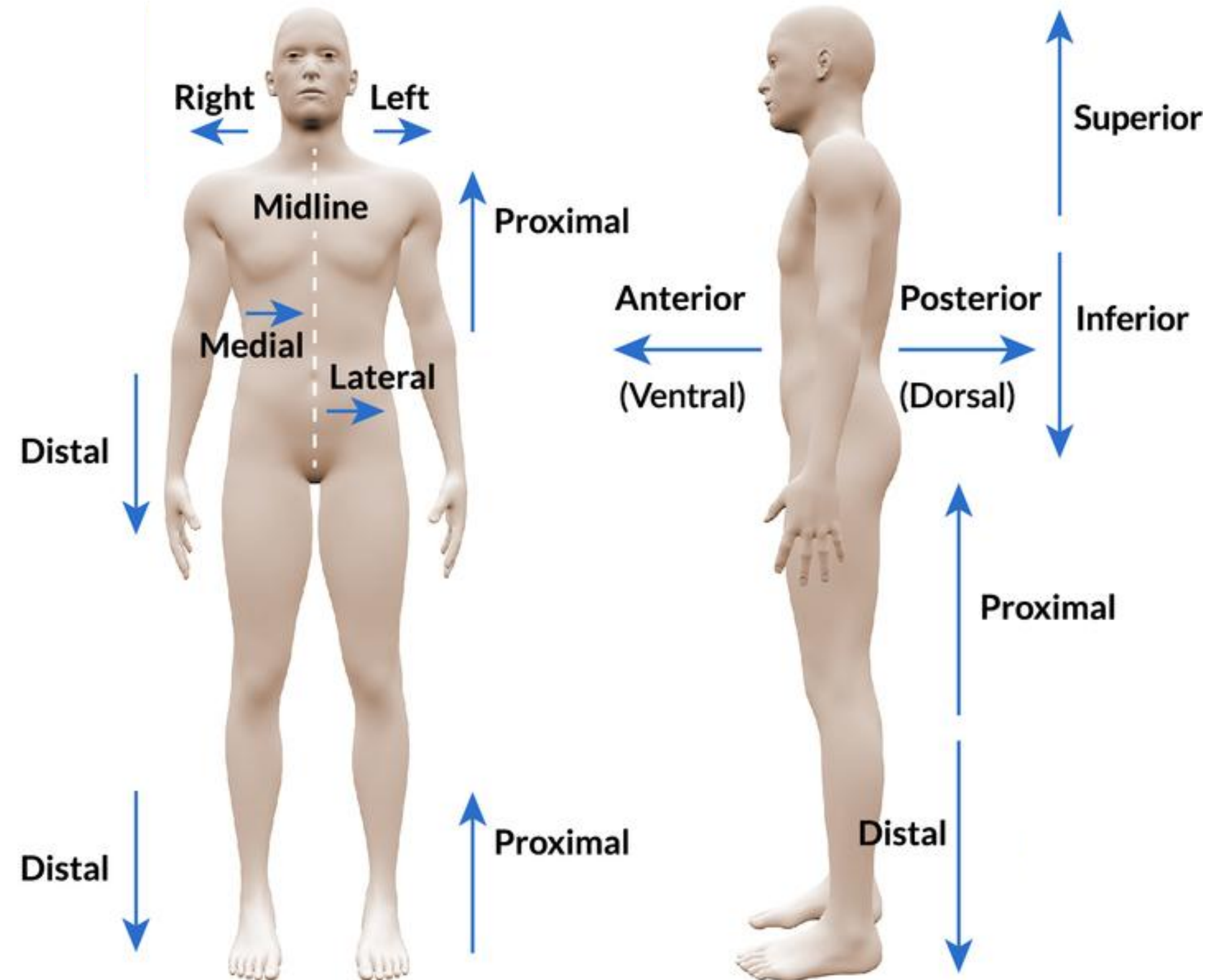
Terms of position (Directional Terms)

- **PROXIMAL:**
nearer to trunk or point of origin (e.g., of a limb).
- **DISTAL:**
farther from trunk or point of origin (e.g., of a limb).



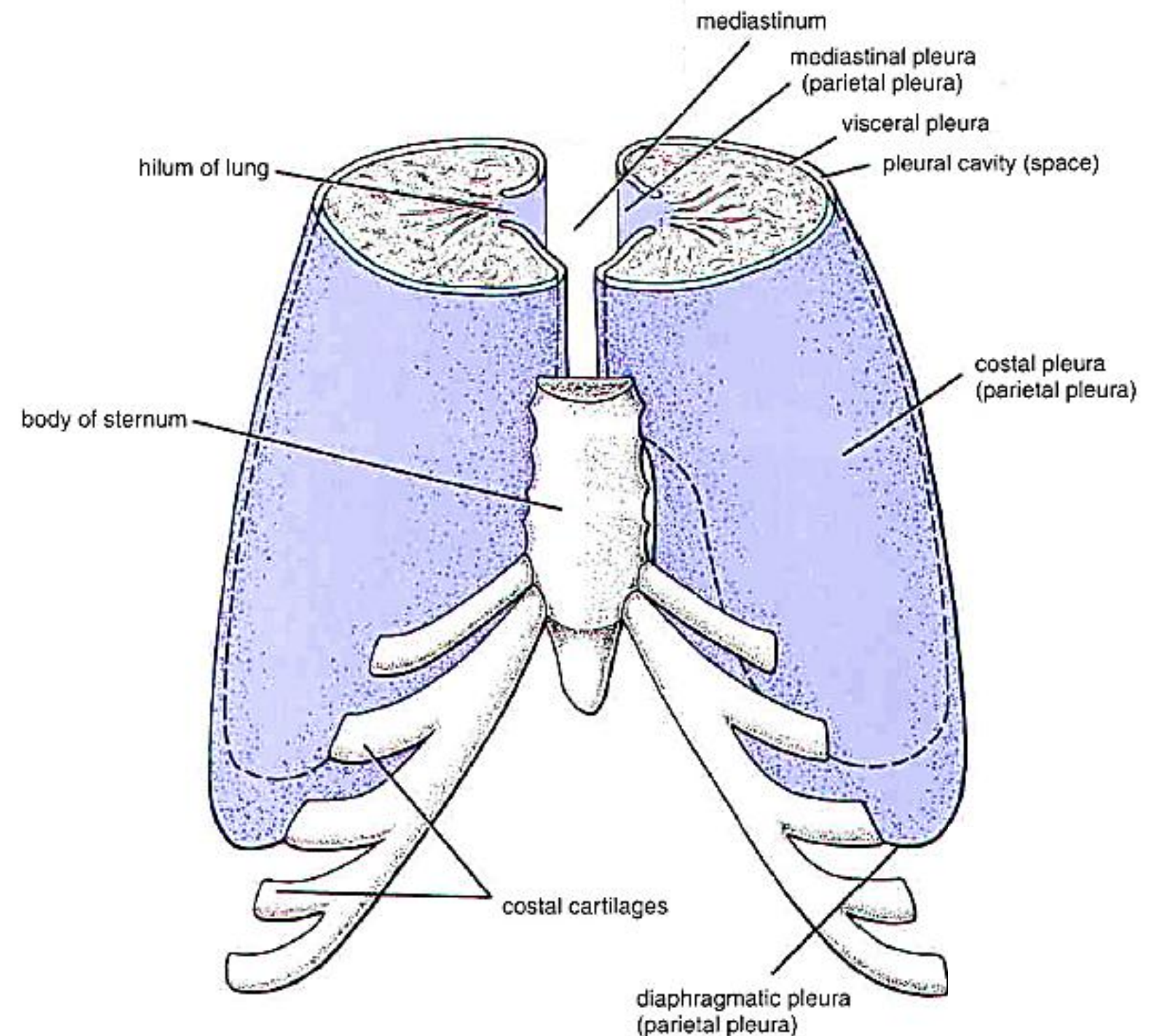
Terms of position (Directional Terms)

- **MEDIAL** :
Nearer to
midline than..
- **LATERAL**:
Farther away
from midline
than...



Terms of position (Directional Terms)

- **INTERNAL** :
inside or closer
to the center,
- **EXTERNAL**:
outside of or
farther from the
center of an
organ or cavity.



Terms of position (Directional Terms)

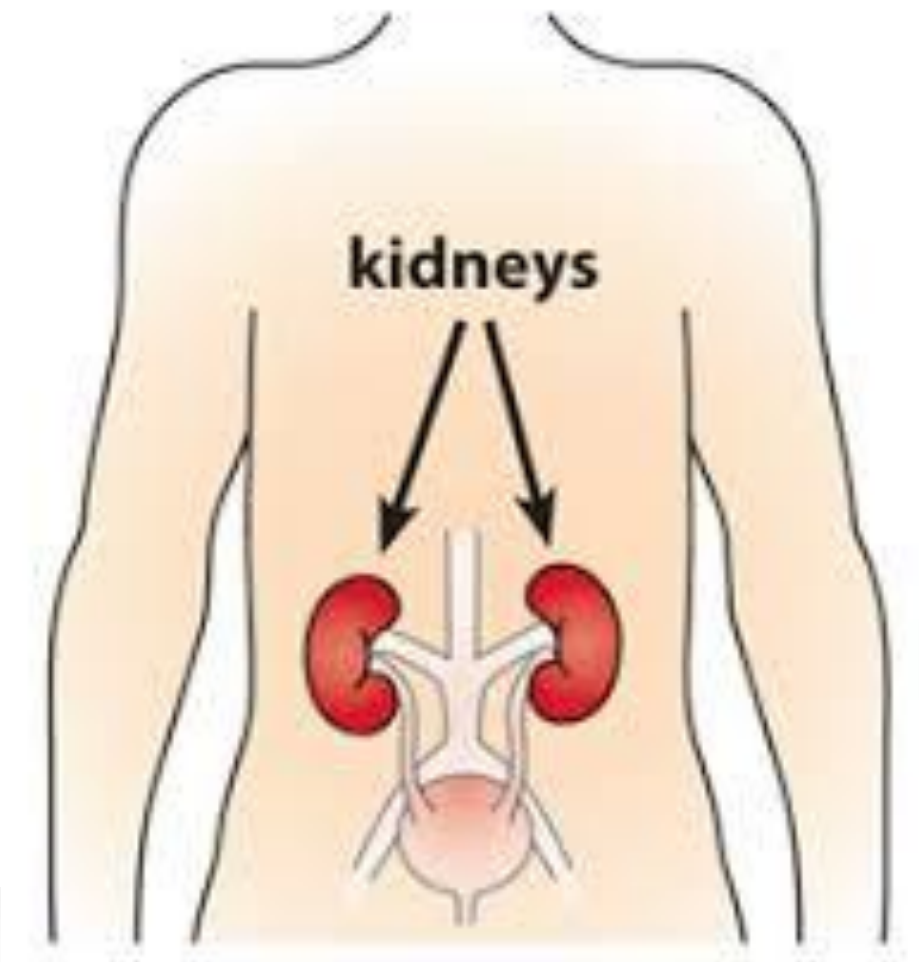
- Superficial
- Intermediate
- Deep

describe the position of structures **relative to the surface of the body**

Terms of laterality

- **BILATERAL:**

Paired structures having right and left members (e.g., the kidneys)



- **UNILATERAL:**

whereas those occurring on one side only (e.g., the spleen)



Terms of laterality

- **IPSILATERAL :**

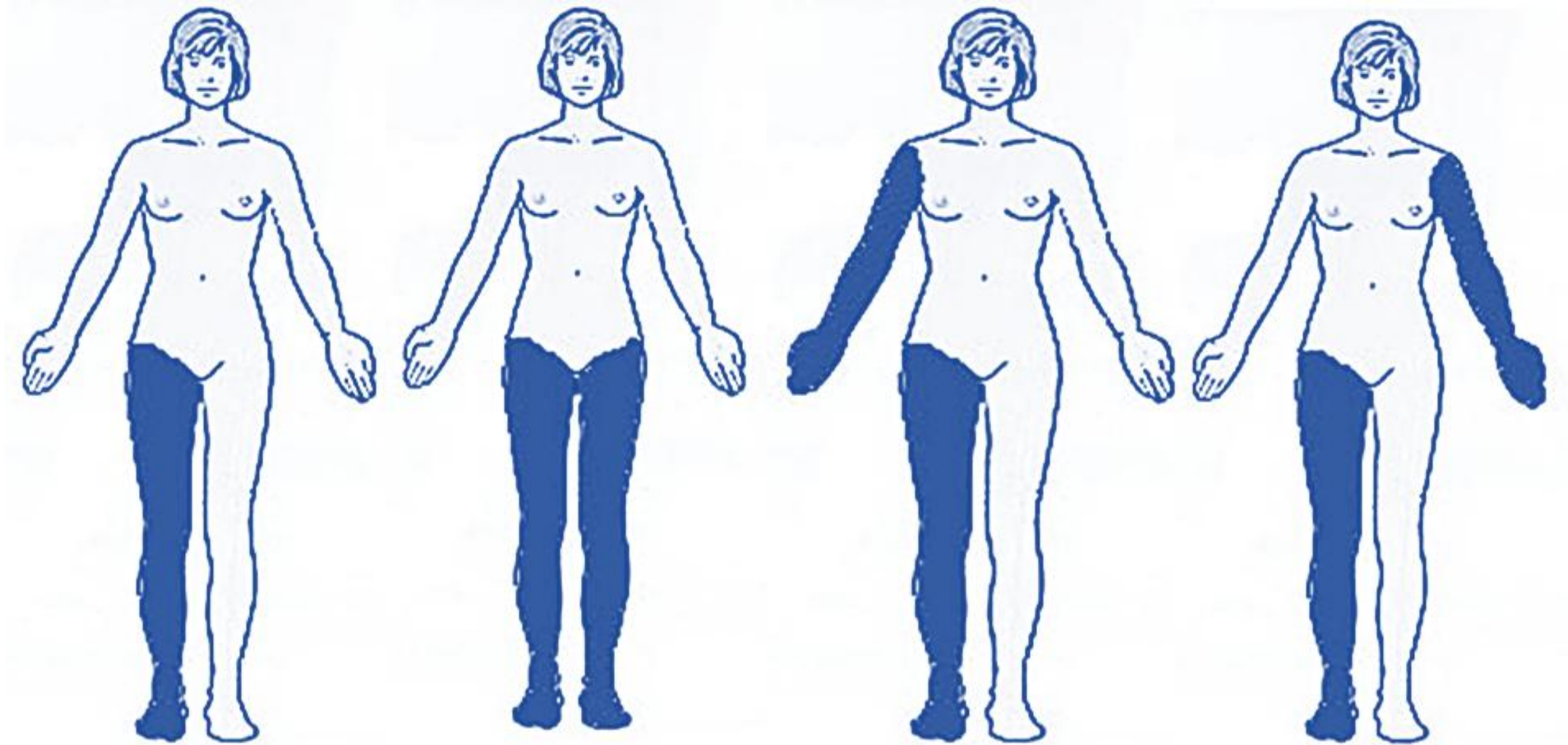
something occurring **on the same side of the body** as another structure is ipsilateral; e.g. the **right thumb** and **right great toe** are ipsilateral.

- **CONTRALATERAL:**

occurring on the **opposite side of the body** relative to **another structure**; the **right hand** is contralateral to the **left hand**.



Terms of laterality



Unilateral:	Bilateral:	Ipsilateral:	Contralateral:
One Side	Two Sides	Same Side of Body	Opposite Side of Body

Terms of position (Directional Terms)

- **SUPINE:**

When the body is lying on the back.

Supine



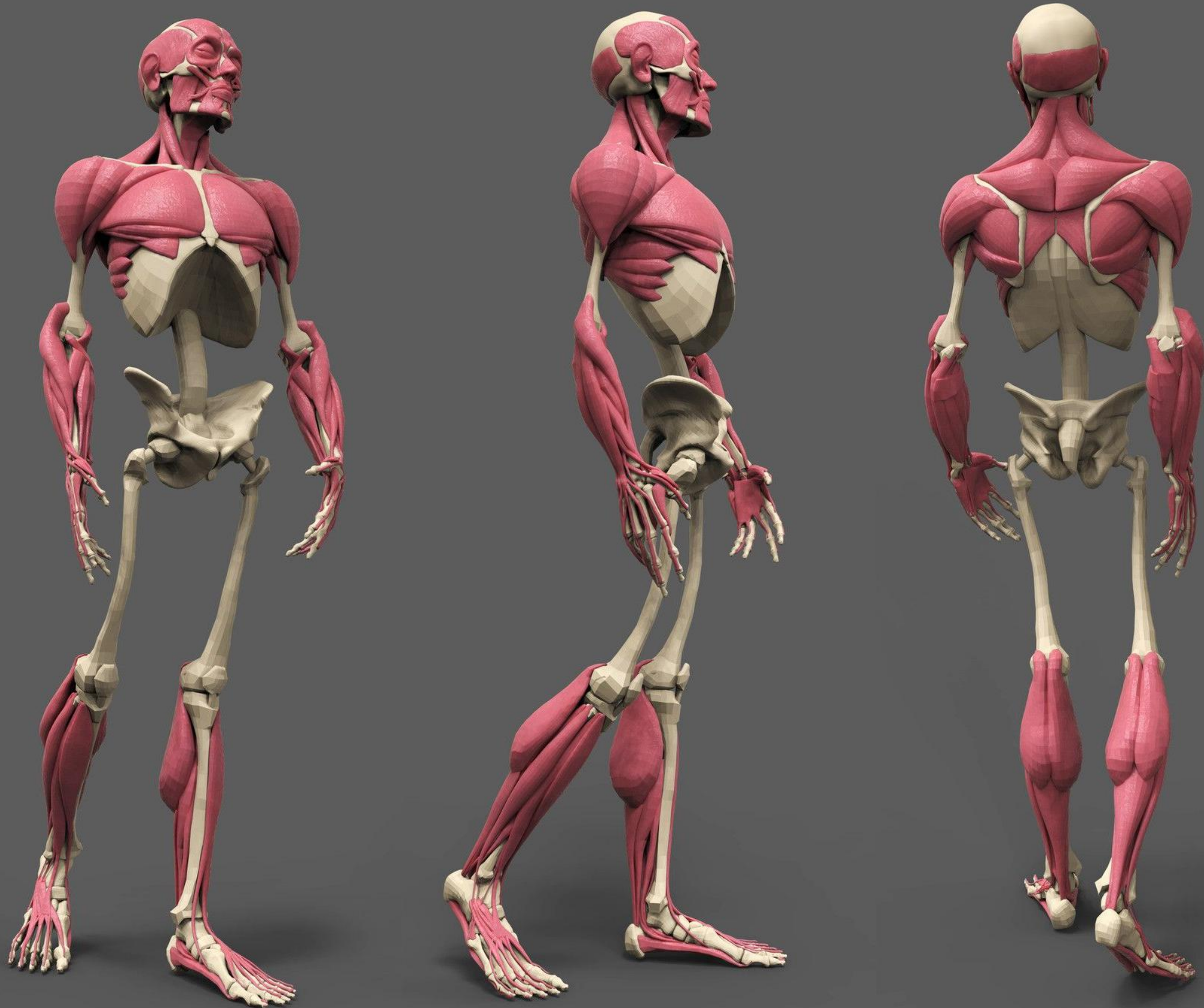
- **PRONE:**

When the body is lying on the face.

Prone



Terms of *Movements*



Joints

- Joints or articulations are unions or junctions **between two or more bones** or rigid parts of the skeleton.



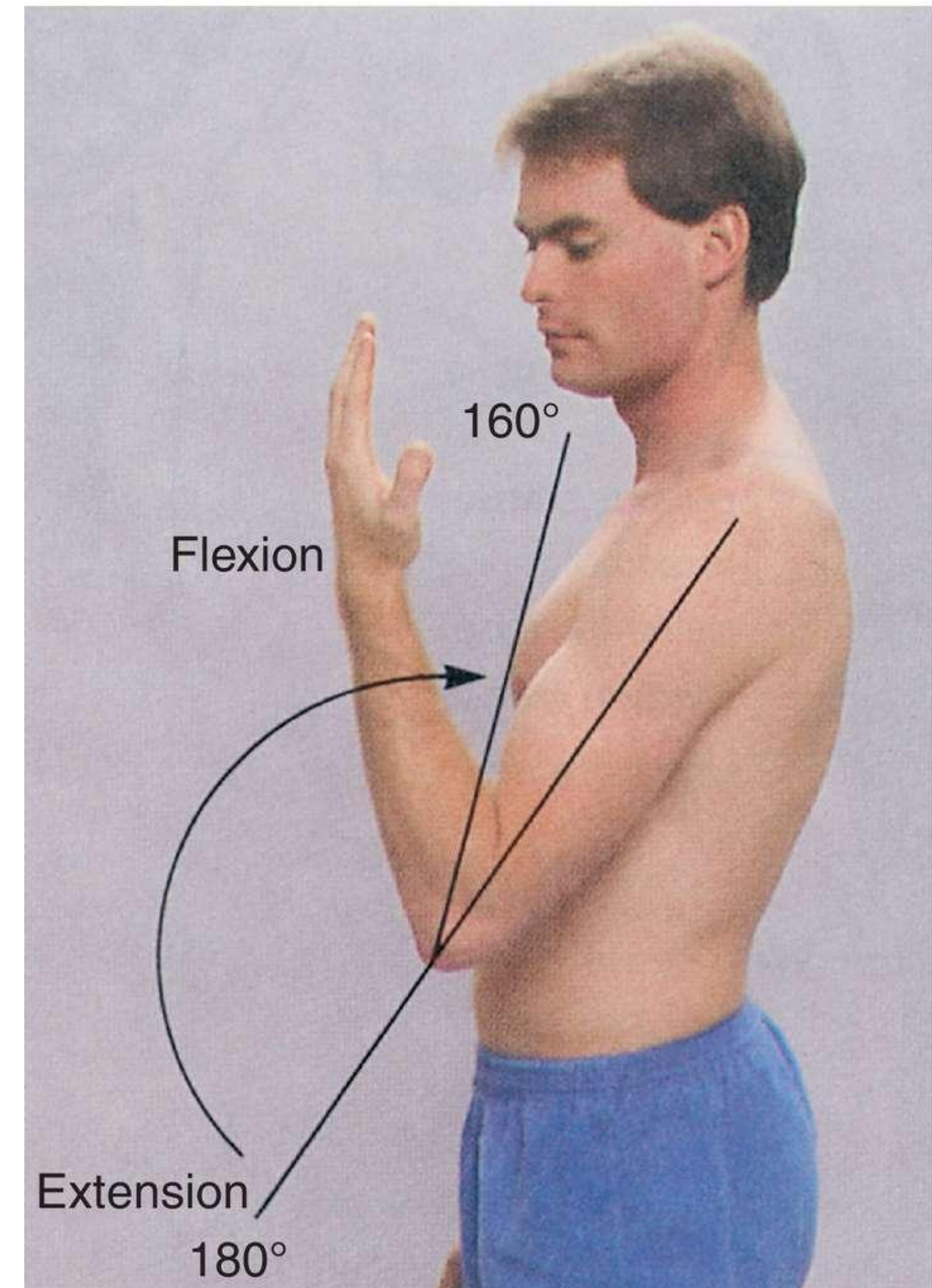
Terms of movement

- **Flexion:**

decreasing the angle between the bones or parts of the body.

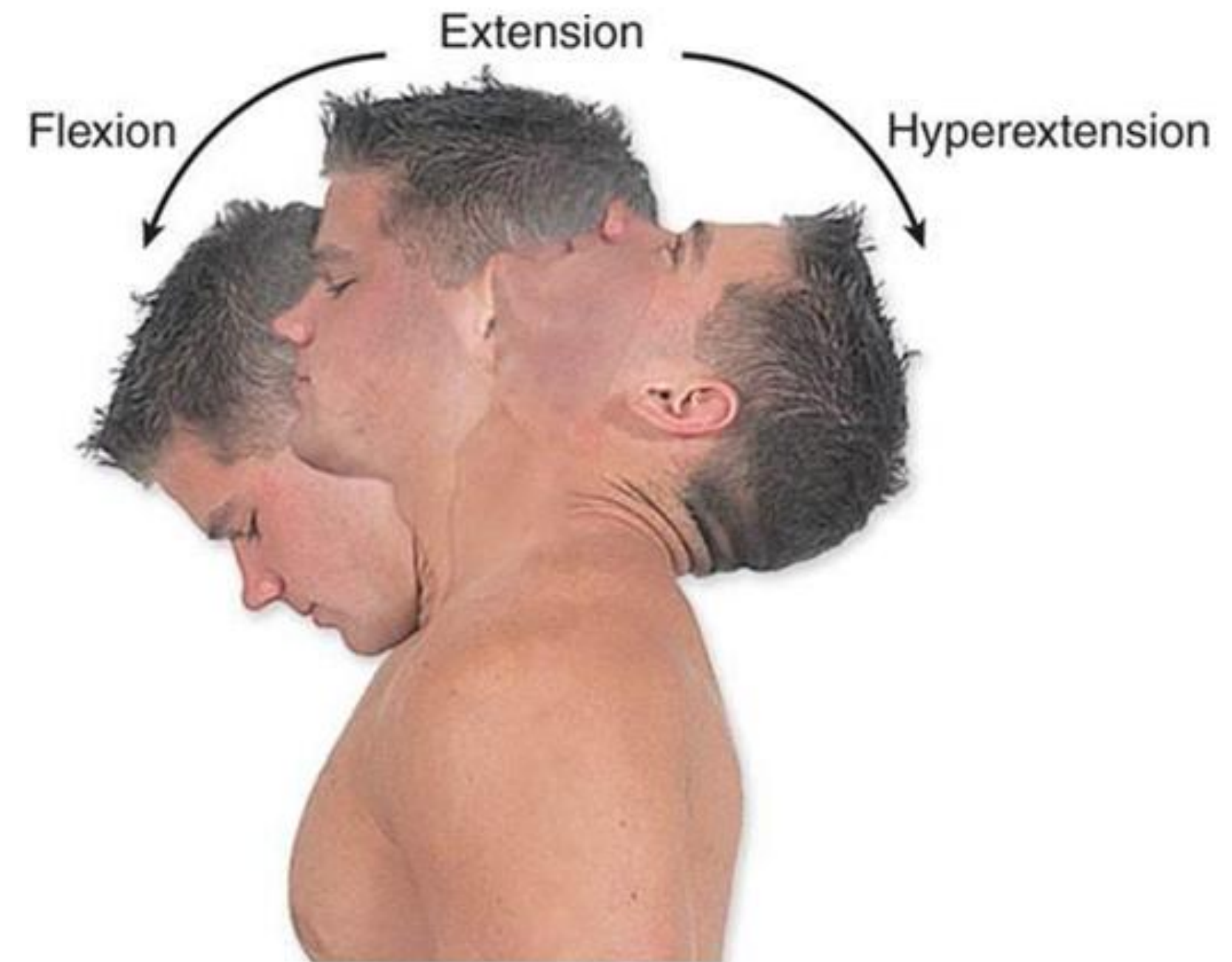
- **Extension:**

indicates straightening or increasing the angle between the bones or parts of the body.



Terms of movement

- Extension of part **beyond the normal limit hyperextension** (overextension).
- Hyperextension **can cause injury**, such as “whiplash” (i.e., hyperextension of the neck during a **rear-end automobile collision**).

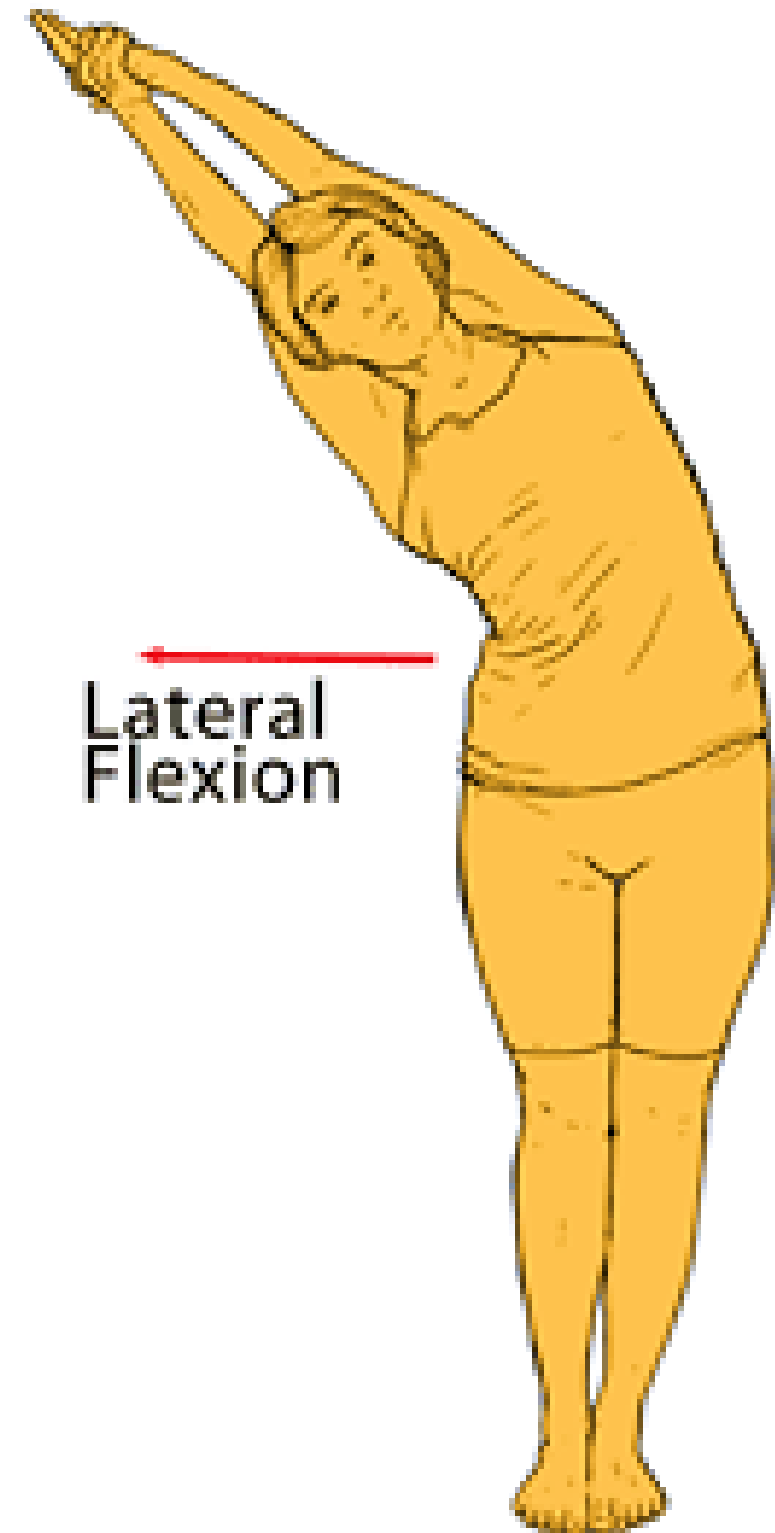


Terms of movement

- **LATERAL FLEXION:**

is a movement of the trunk in the coronal plane

Right and left lateral flexion are **special forms of abduction for only the neck and trunk.**



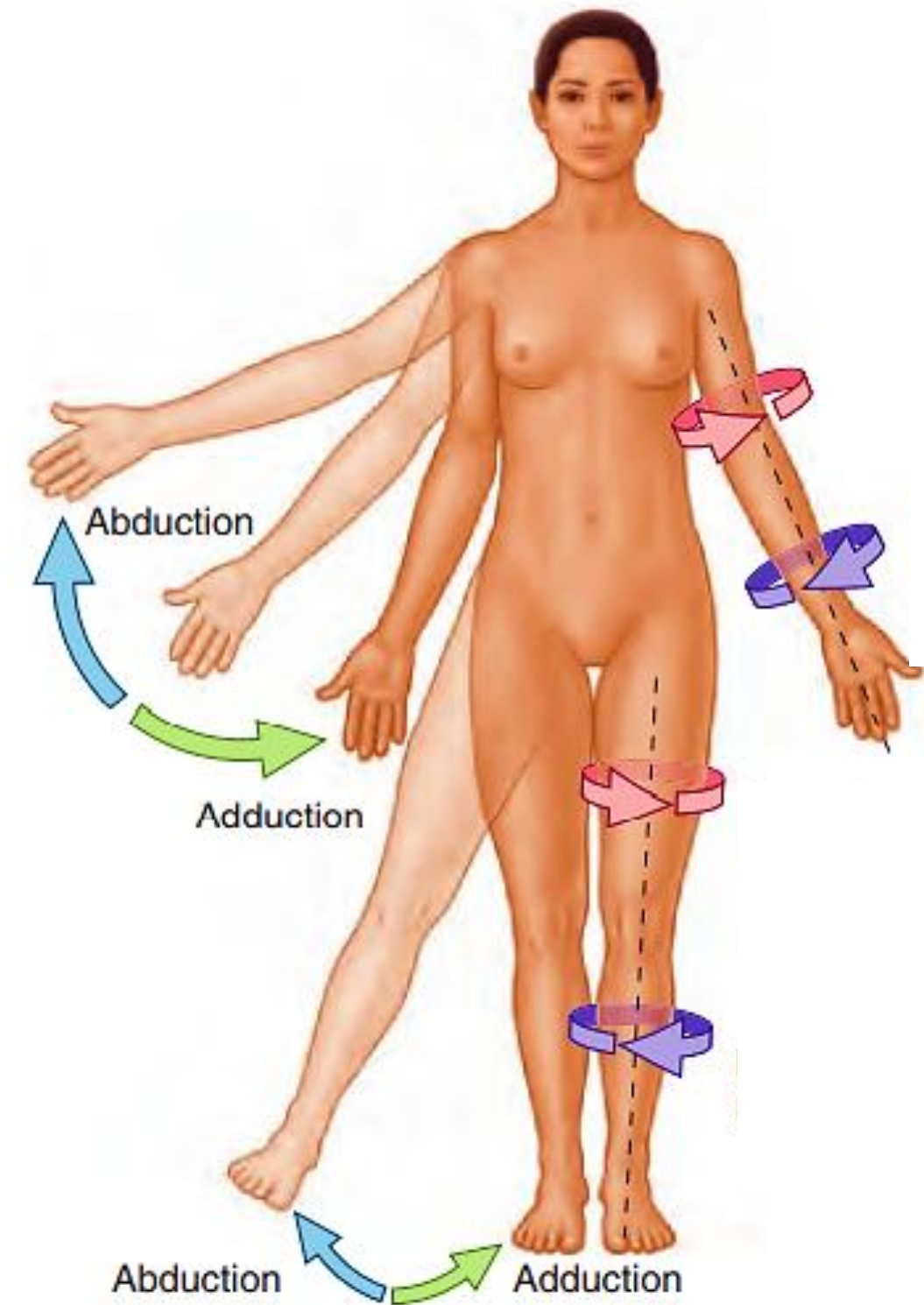
Terms of movement

▪ ABDUCTION:

Movement of a limb in coronal plane away from midline of body.

▪ ADDUCTION:

Movement of a limb in coronal plane towards the body.

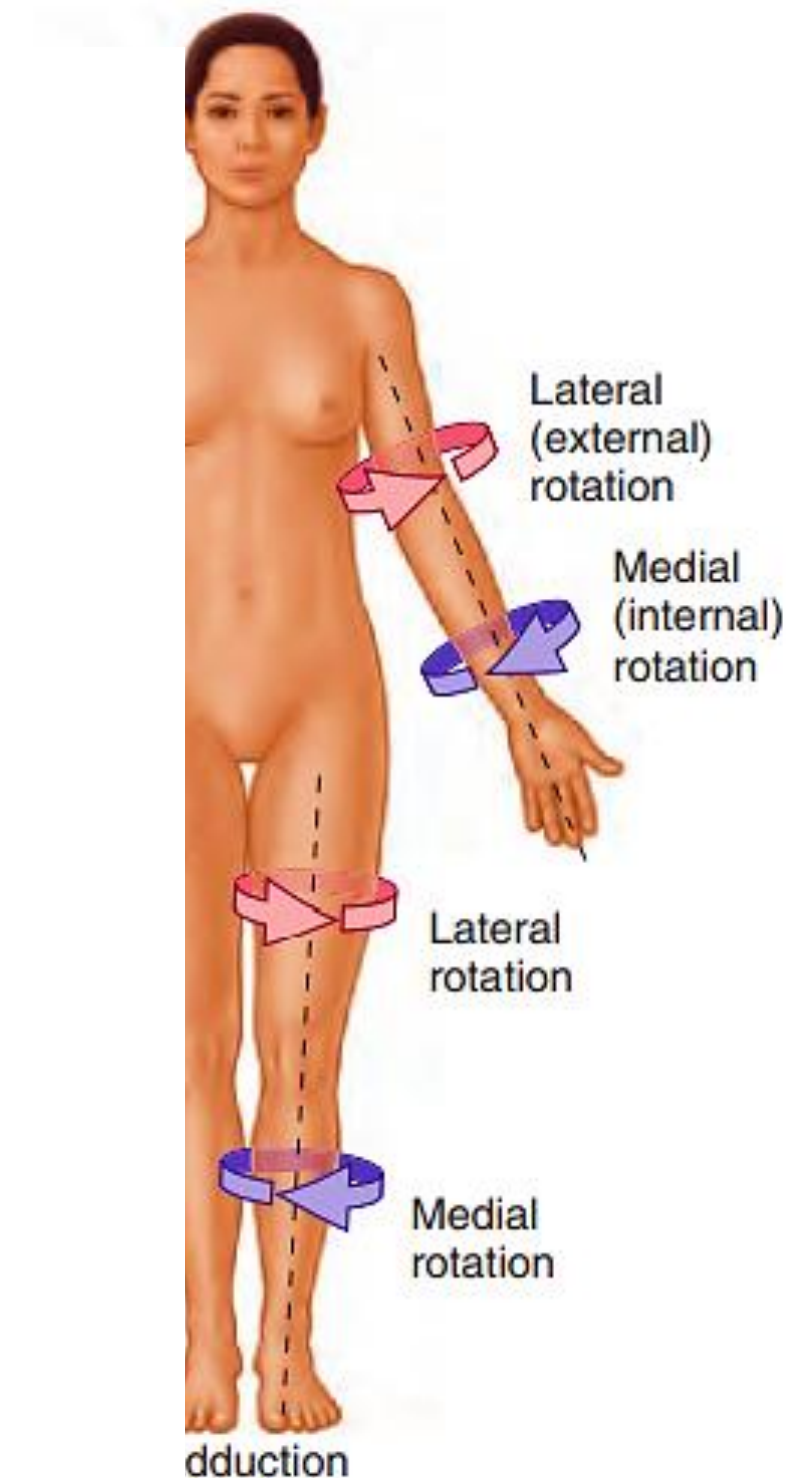


(G) Abduction and adduction of right limbs and rotation of left limbs at glenohumeral and hip joints, respectively

Terms of movement

■ Rotation :

Movement of a part of the body around its long axis



(G) Abduction and adduction of right limbs and rotation of left limbs at glenohumeral and hip joints, respectively

Terms of movement

- **MEDIAL ROTATION:**

Anterior surface facing medially.



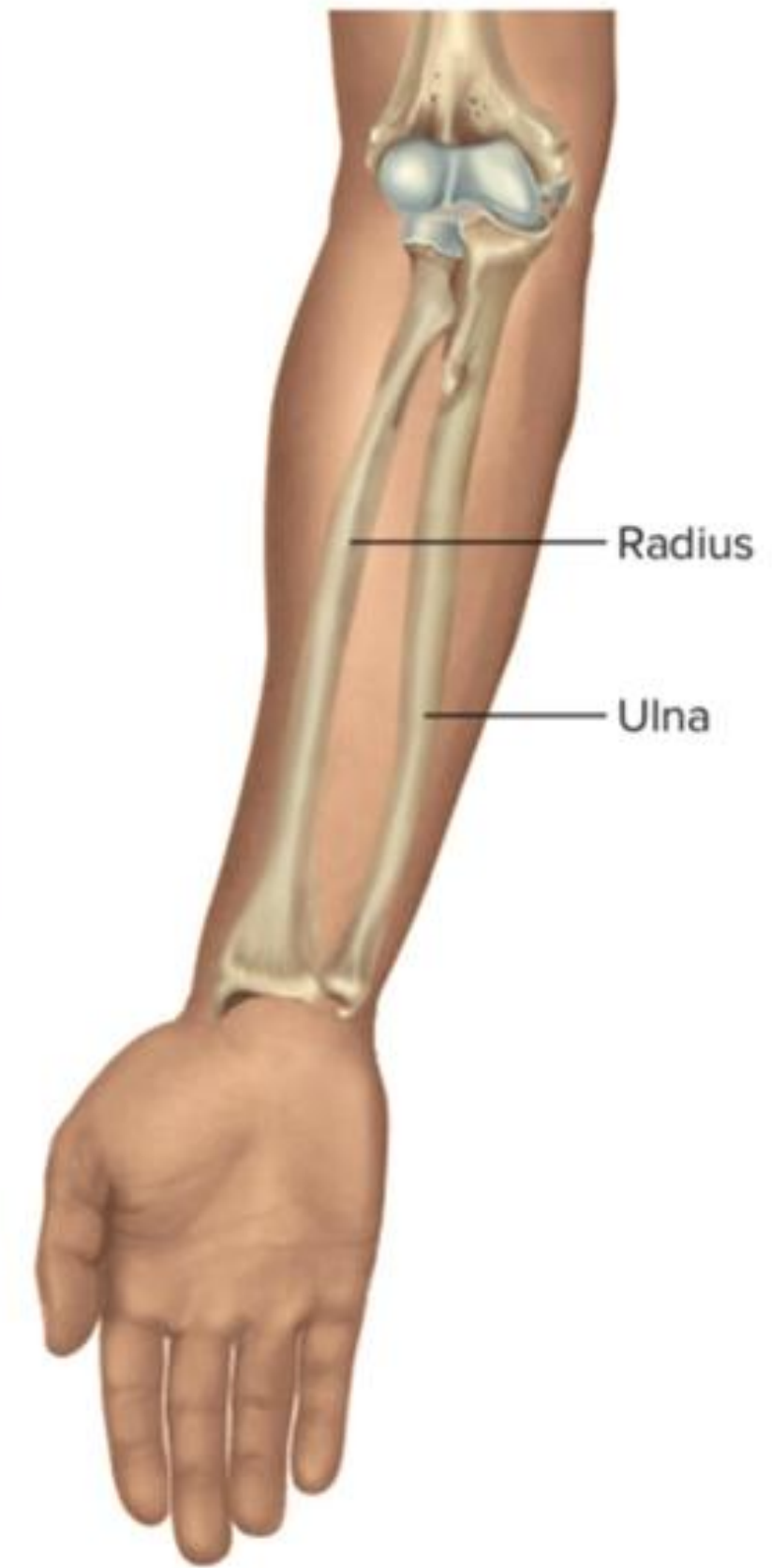
- **LATERAL ROTATION:**

Anterior surface facing laterally.



Terms of movement

- **SUPINATION:**
lateral rotation of a
pronated forearm, so the
**palm of the hand faces
anteriorly.**

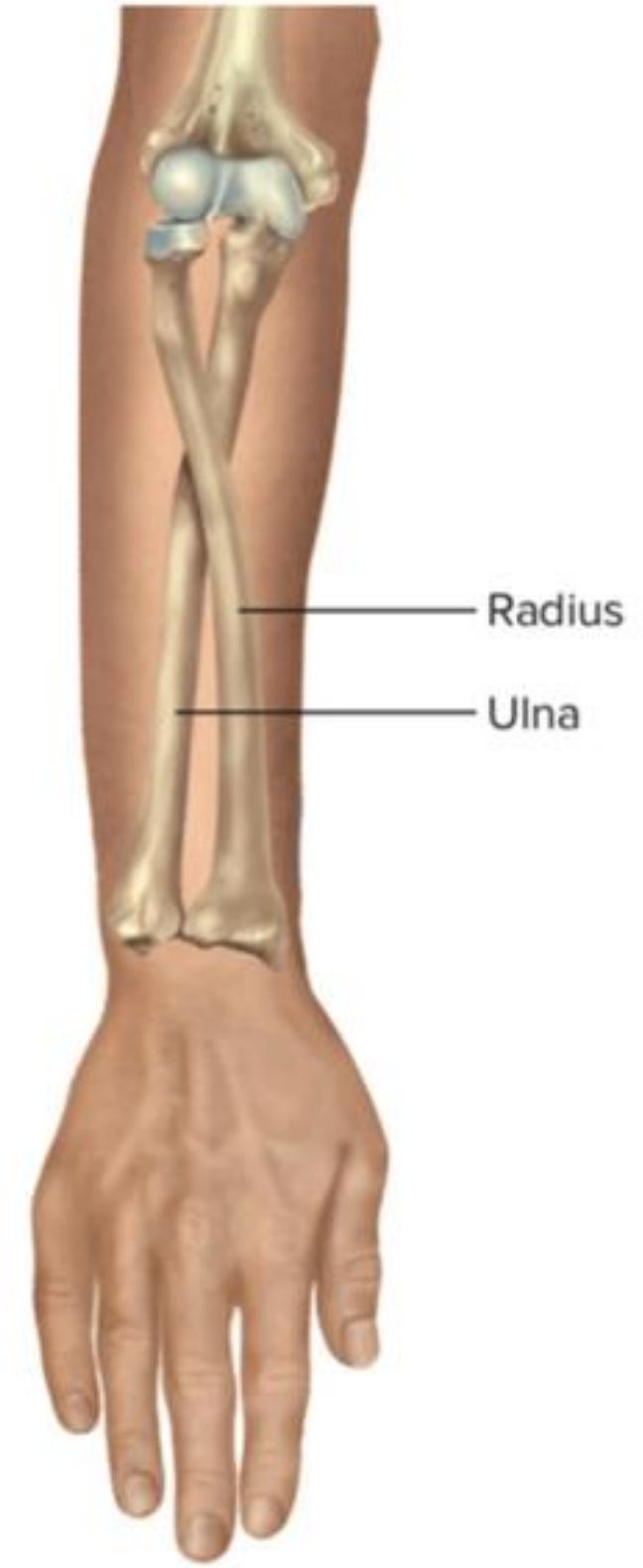


(a) Supination

Terms of movement

■ PRONATION:

medial rotation of forearm so **the palm of the hand faces posteriorly.**



(b) Pronation

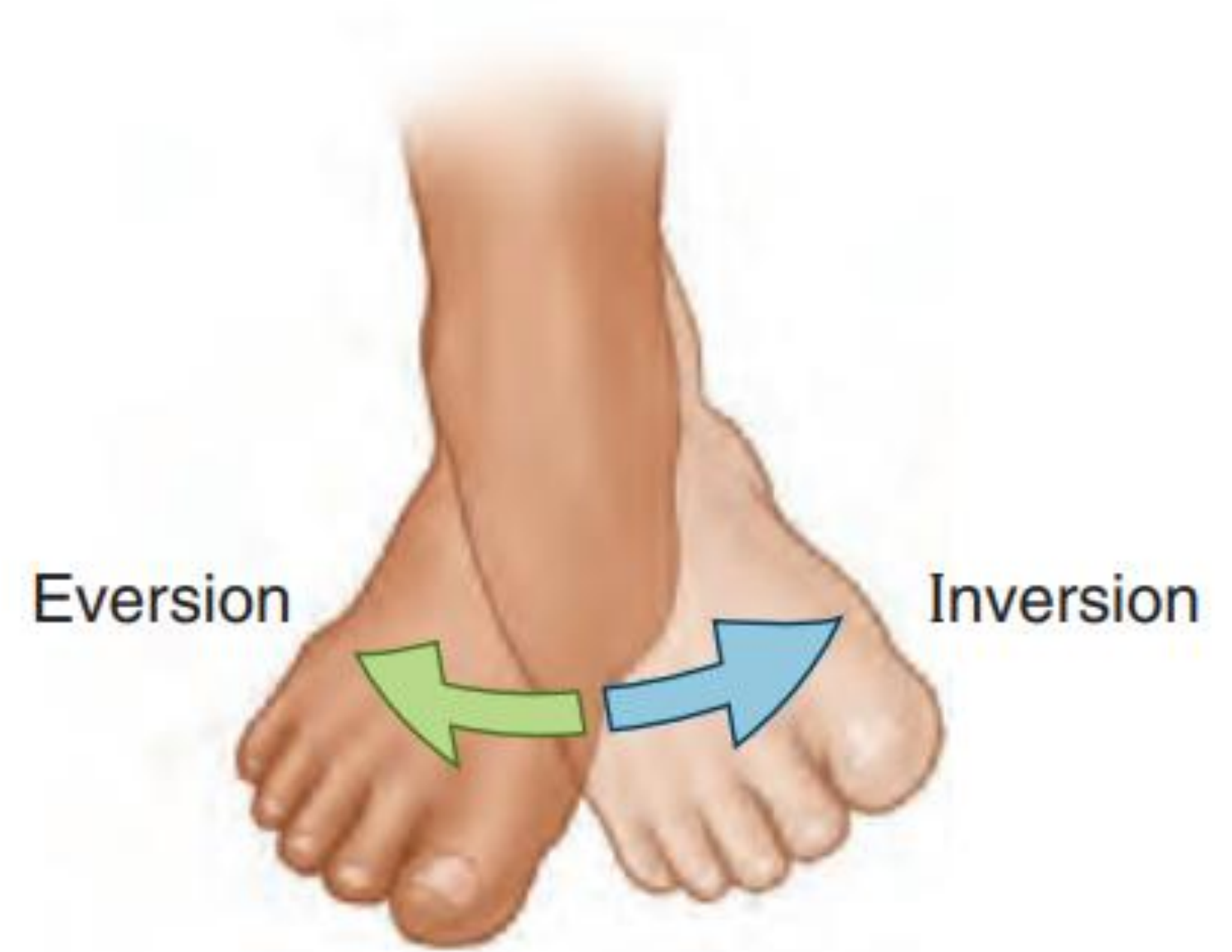
Terms of movement

- **EVERSION:**

movement of the foot so that **the sole faces laterally**.

- **INVERSION:**

movement of the foot so **the sole faces medially**.

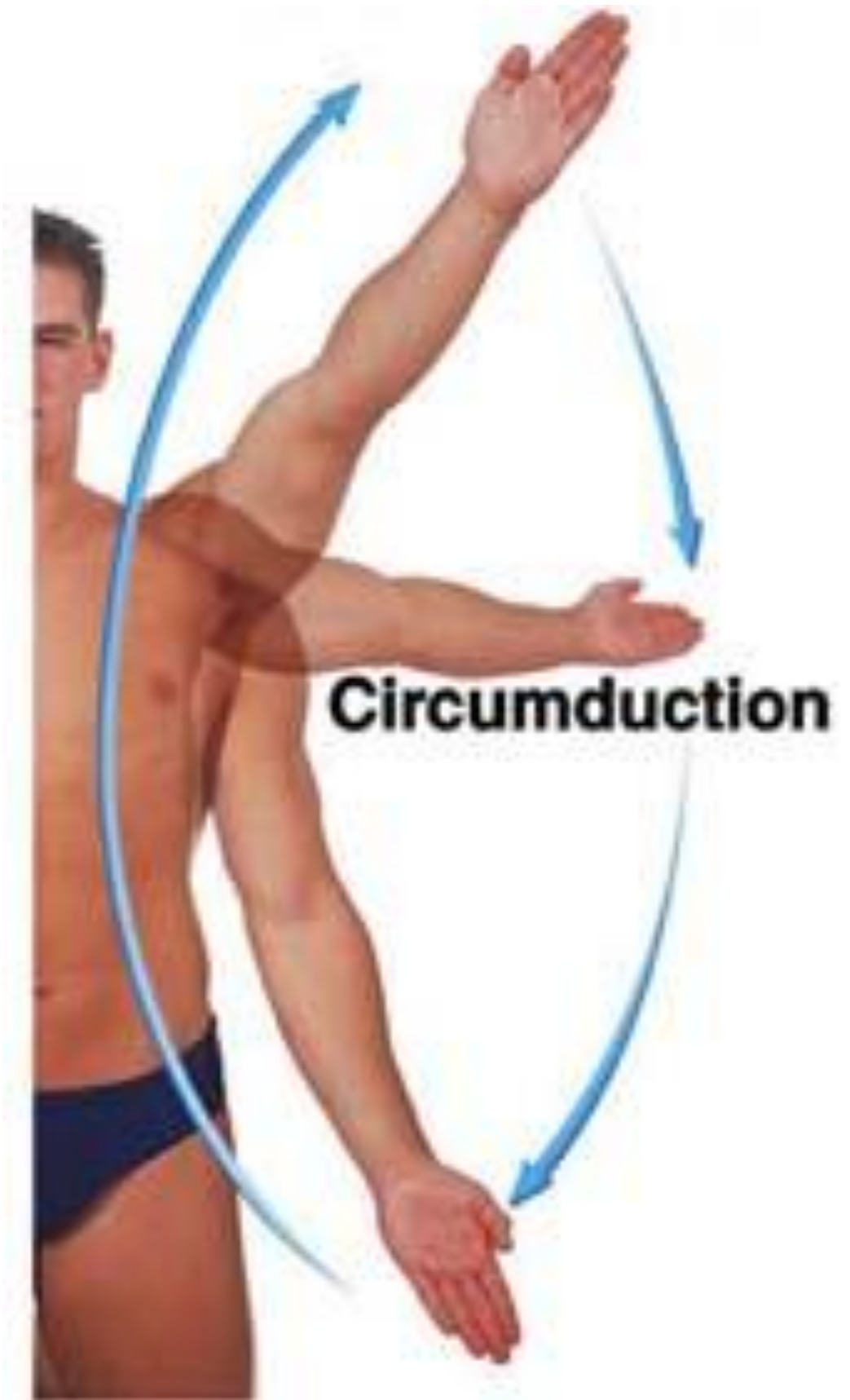


(I) Inversion and eversion of foot at subtalar and transverse tarsal joints

Terms of movement

▪ CIRCUMDUCTION:

- The circular (or, more precisely, conical) **movement of a body part**, such as a ball-and-socket joint or the eye.
- It consists of a **combination** of flexion, extension, adduction, and abduction.
- A rotatory movement of joints



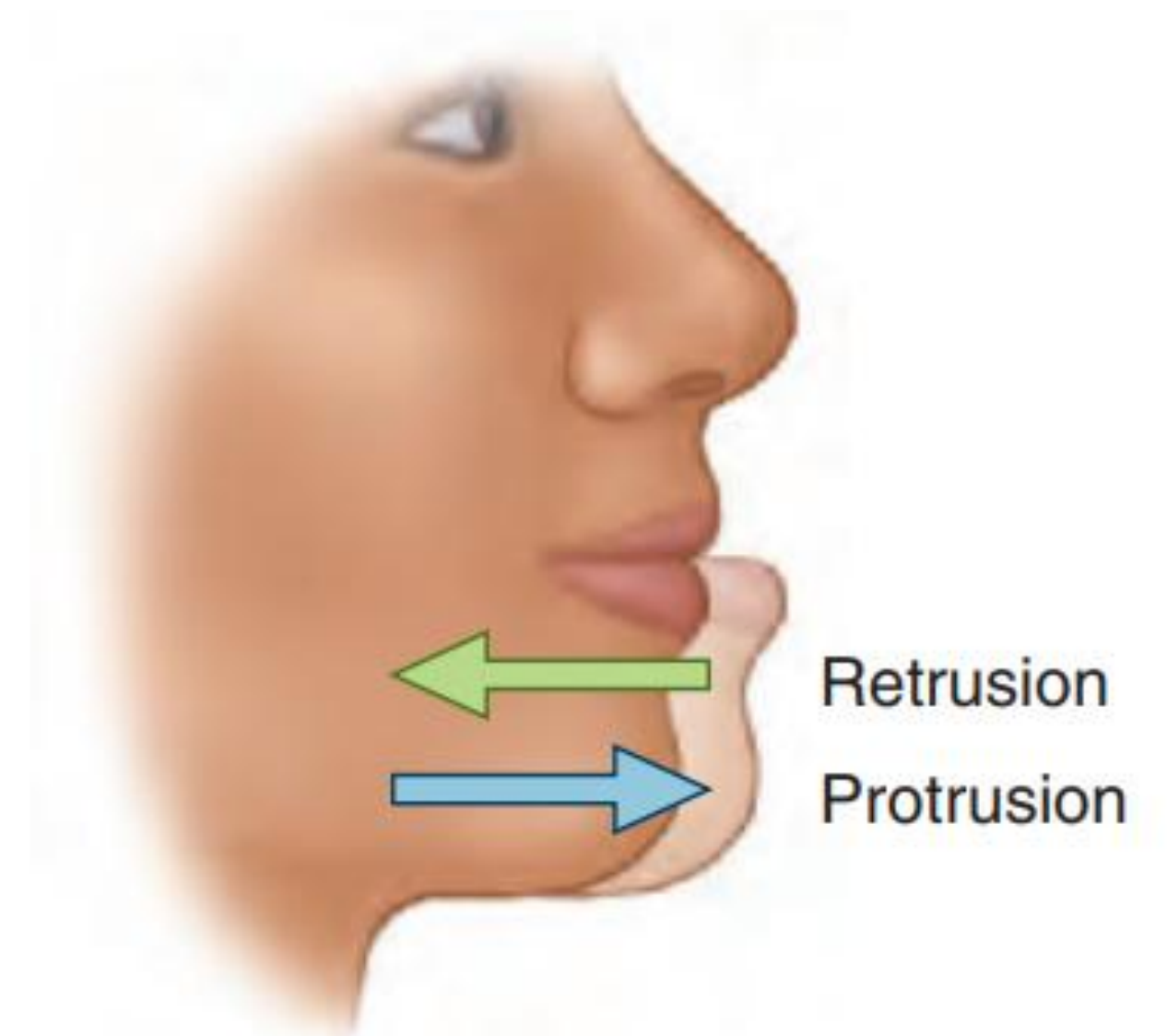
Terms of movement

■ PROTRUSION:

is a movement anteriorly (forward) as in protruding the mandible, lips, or tongue.

■ RETRUSION:

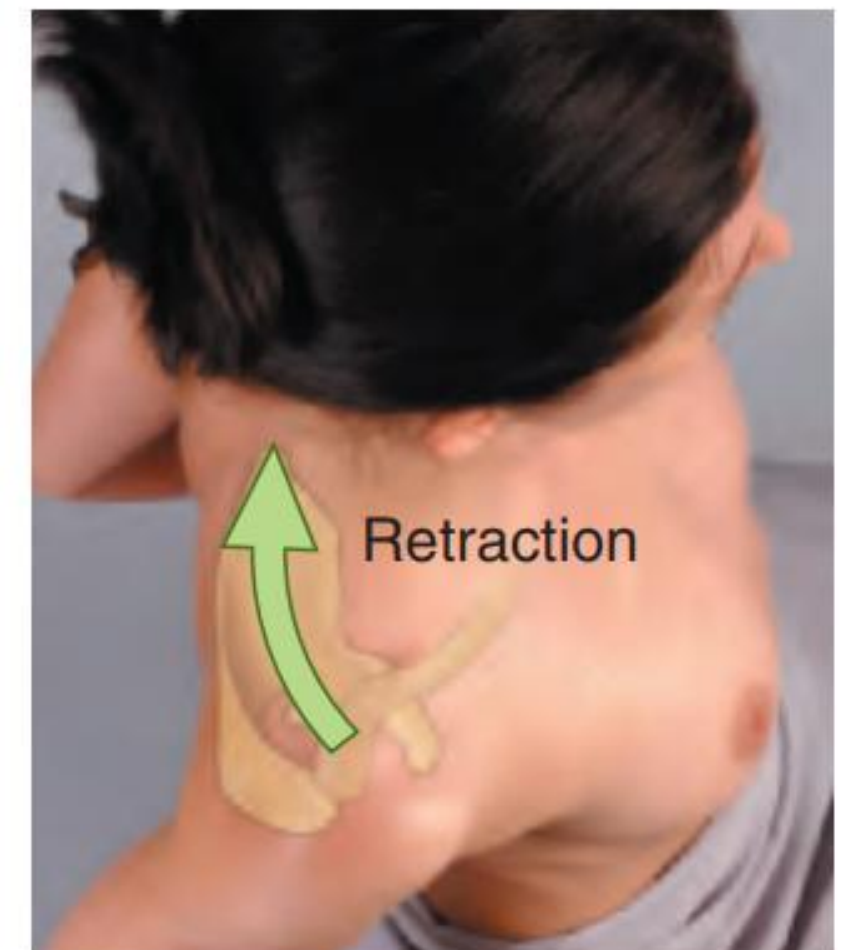
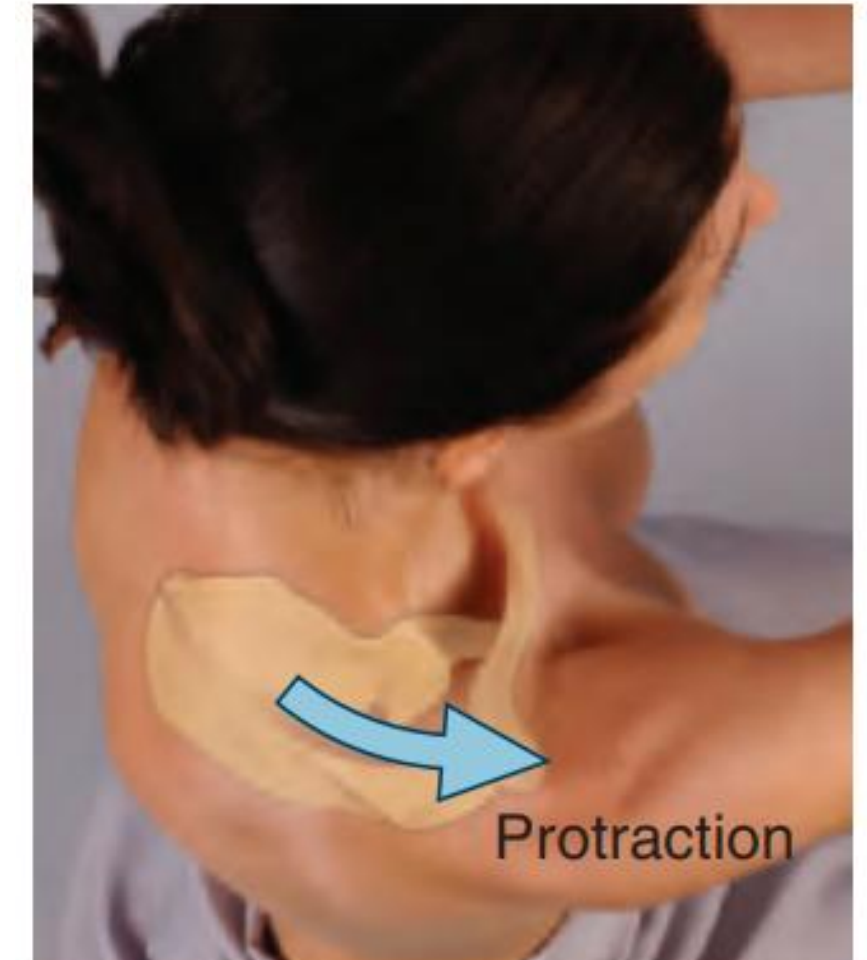
is a movement posteriorly (backward), as in retruding the mandible, lips, or tongue.



(L) Protrusion and retrusion of jaw at temporomandibular joints

Terms of movement

- The similar terms **protraction** and **retraction** are used for **anterolateral** and **posteromedial** movements of the **scapula** on the thoracic wall, causing the shoulder region to move **anteriorly** and **posteriorly**.



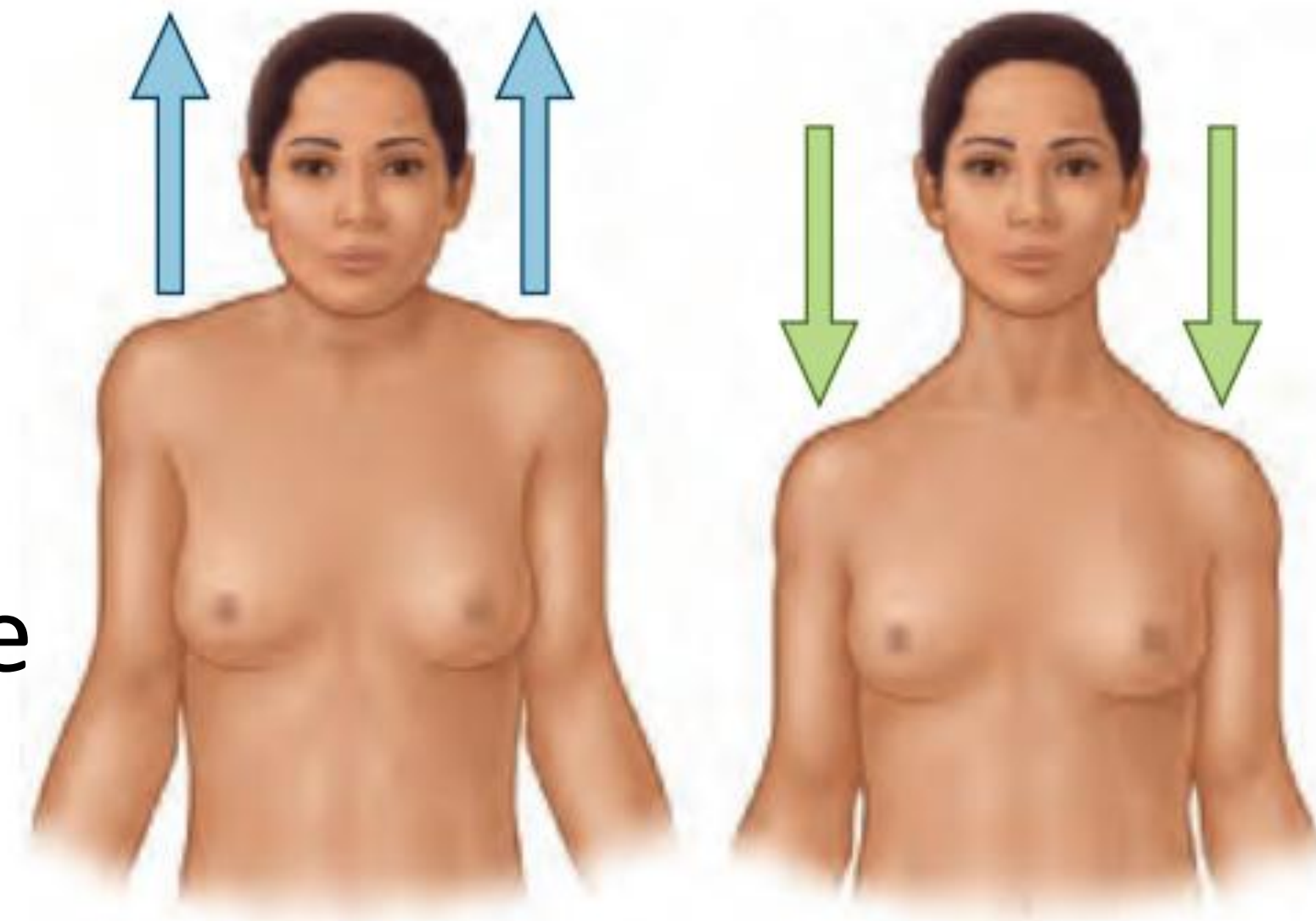
Terms of movement

- **ELEVATION:**

raises or moves a part superiorly,
>as in elevating the shoulders,
>the upper eyelid when opening the eye.

- **DEPRESSION:**

lowers or moves a part inferiorly.



Elevation

Depression

(K) Elevation and depression
of shoulders

Terms of movement

■ DORSIFLEXION:

flexion at the ankle joint, as occurs when walking uphill or **lifting the front of the foot and toes off the ground.**

■ PLANTARFLEXION:

bends the foot and toes toward the ground, as when **standing on your toes.**



Terms of movement

- **Radial Deviation:**

Movement of the wrist towards the radius or lateral side.

- **Ulnar Deviation:**

Movement of the wrist towards the ulna or medial side.



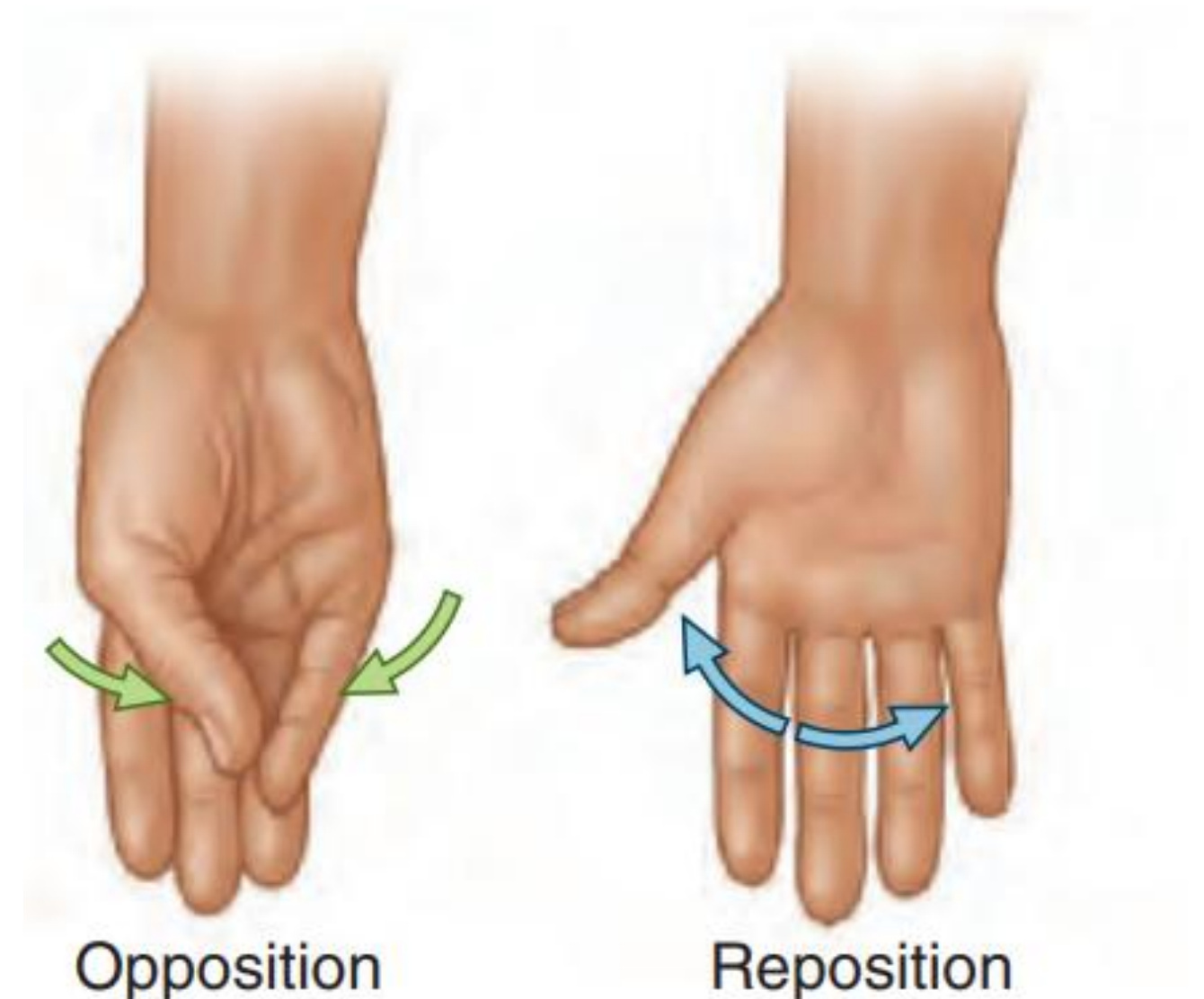
Terms of movement

- **Opposition:**

Movement of the thumb across the palm of the hand is brought to another digit pad.

- **Reposition:**

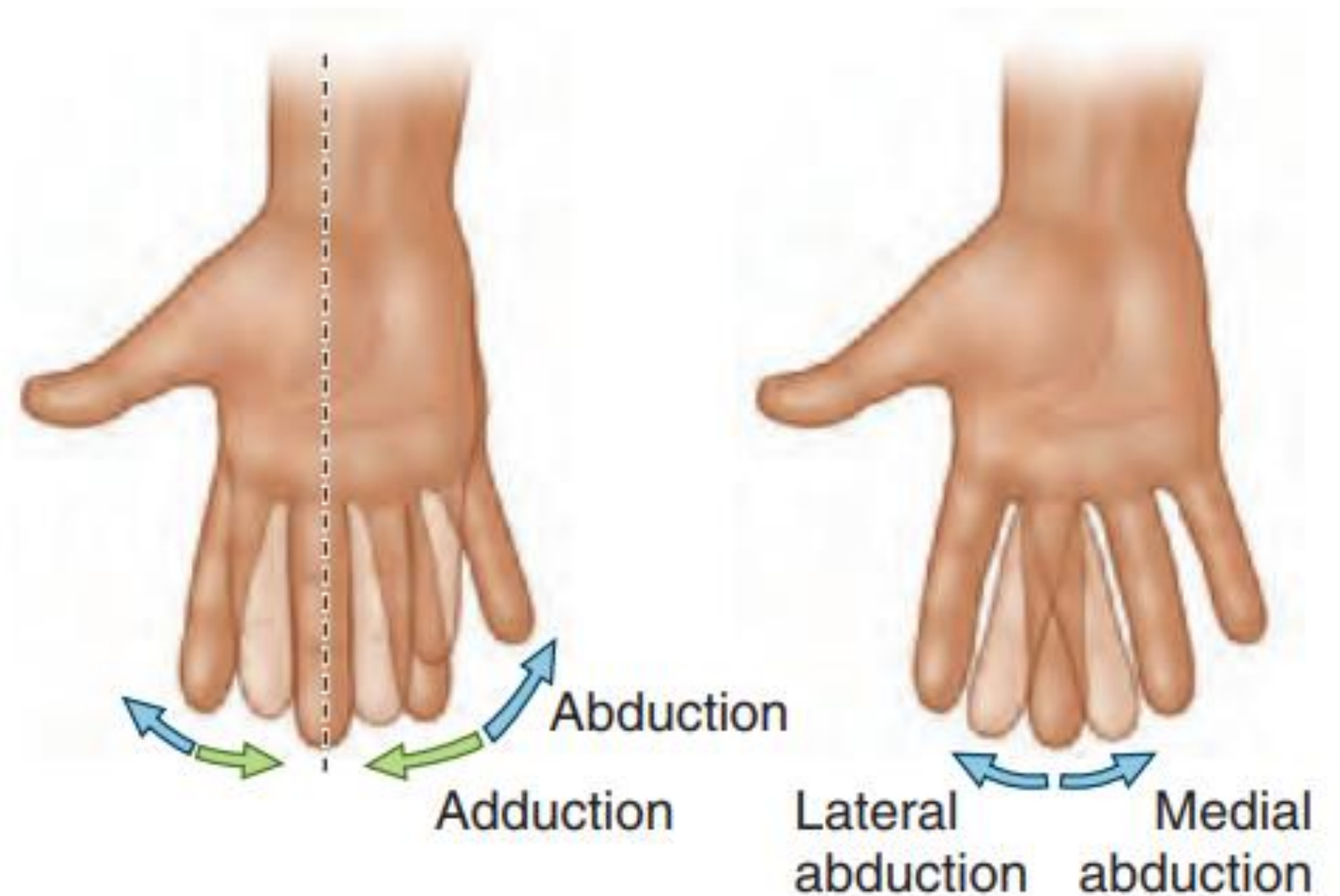
describes the movement of the 1st digit from the position of opposition back to its anatomical position.



Opposition and reposition of thumb and little finger at carpometacarpal joint of thumb combined with flexion at metacarpophalangeal joints

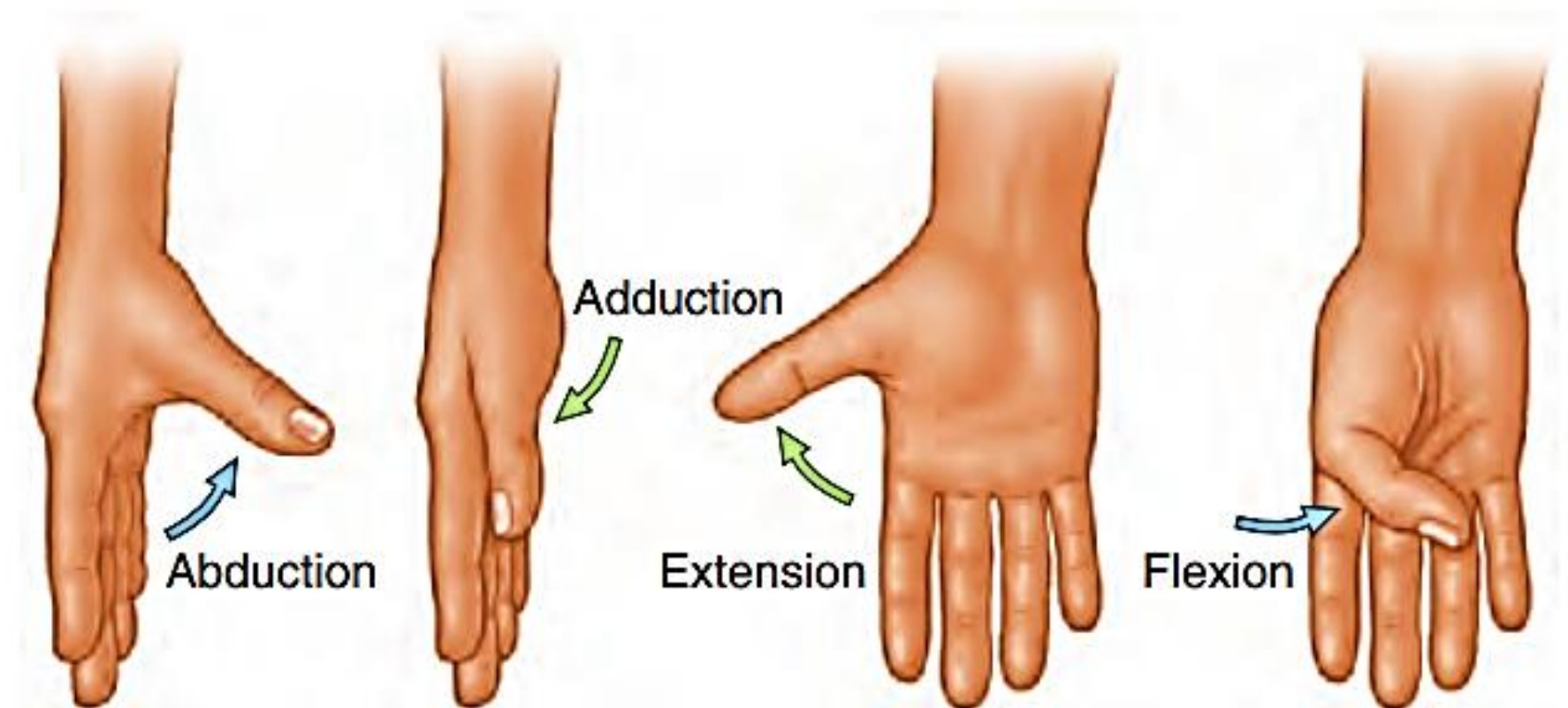
Exceptional movements terms

- **Abduction of the digits** means spreading them apart.
- **Adduction of the digits** is the opposite—**bringing** the spread fingers or toes together.

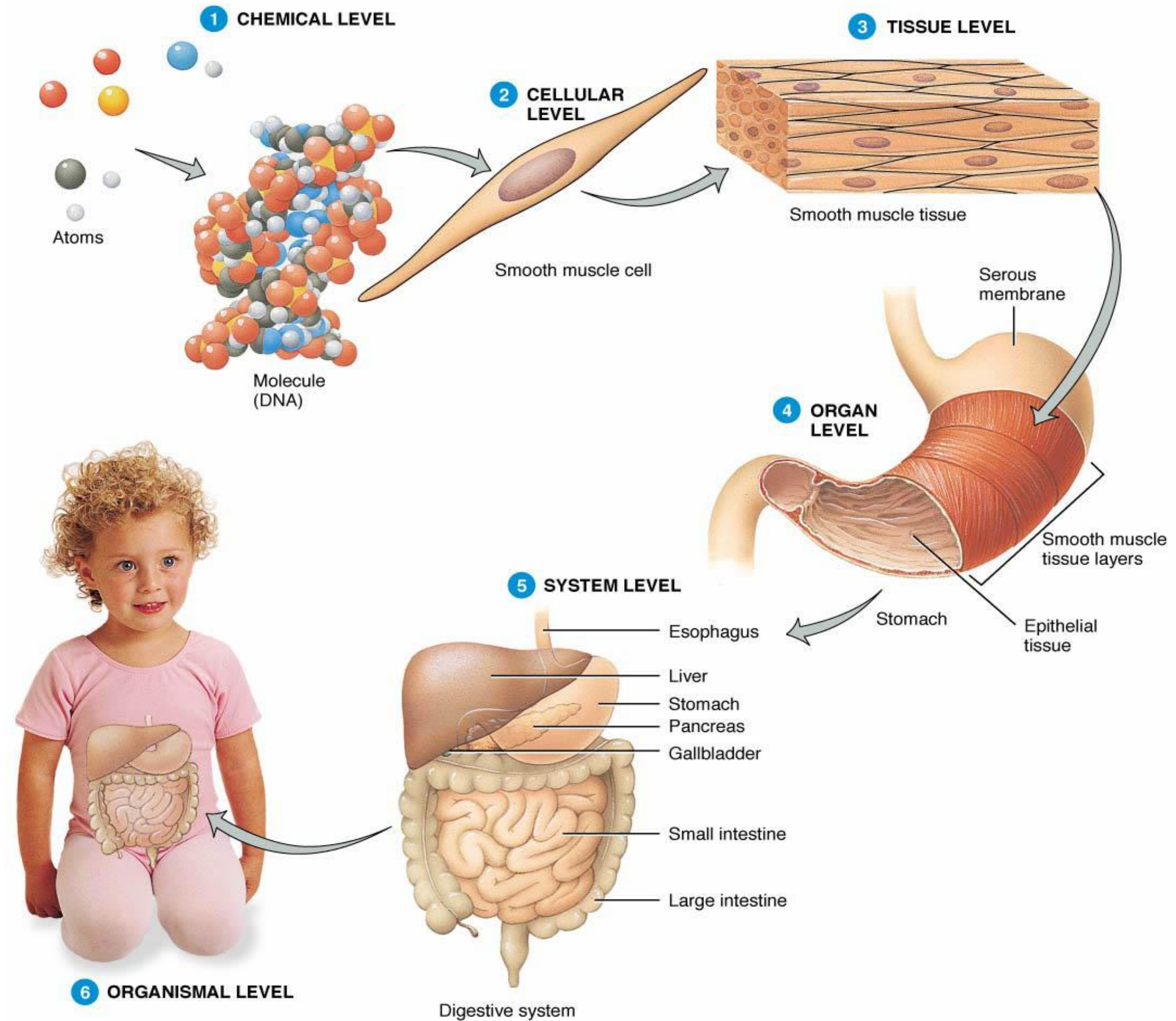


Exceptional movements terms

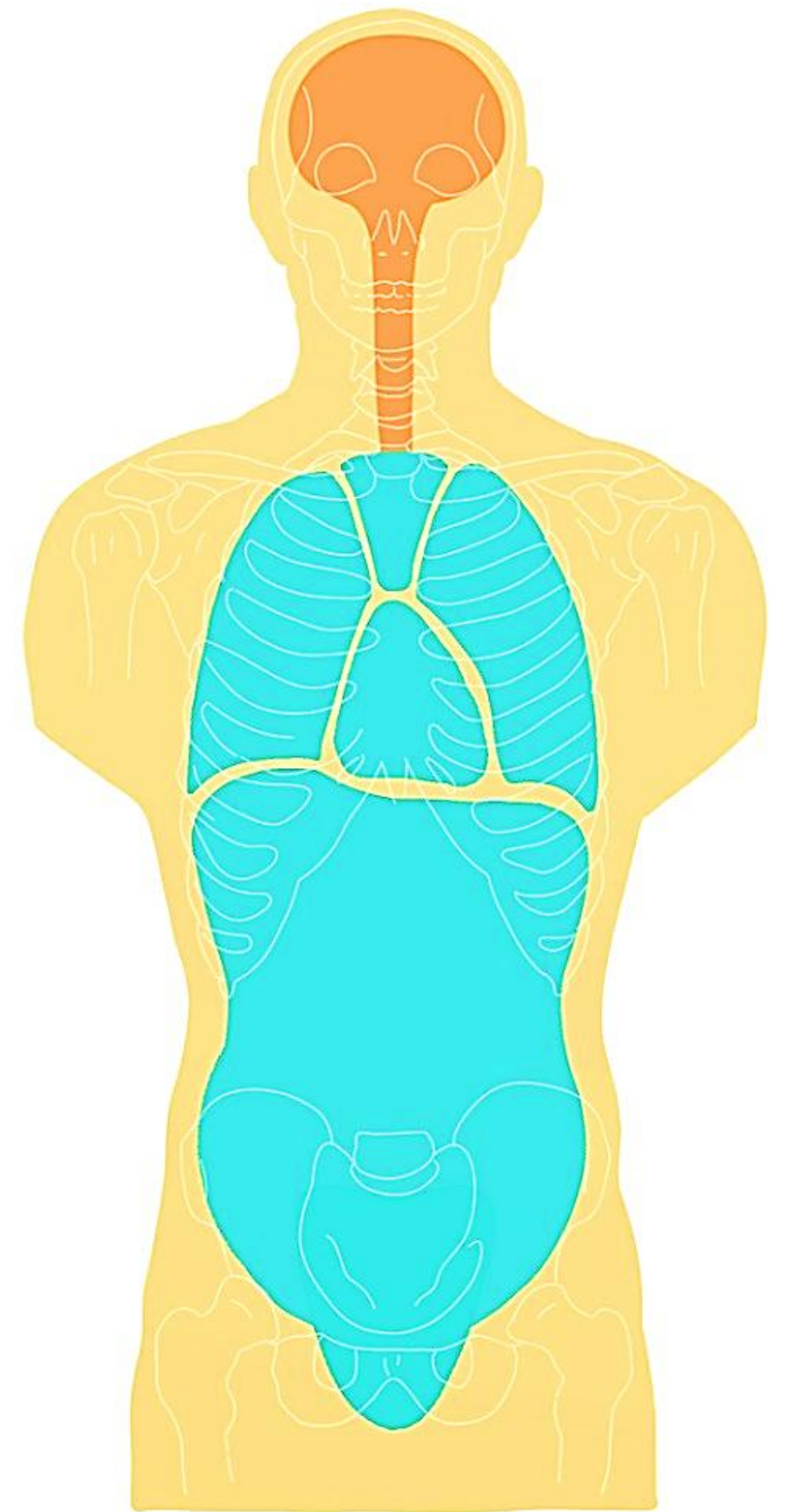
- **The thumbnail of the thumb faces** (laterally instead of posteriorly in the anatomical position),
- The thumb is rotated 90° relative to the other digits.
- Therefore, **the thumb flexes and extends in the coronal plane and abducts and adducts in the sagittal plane.**



Levels of body organization

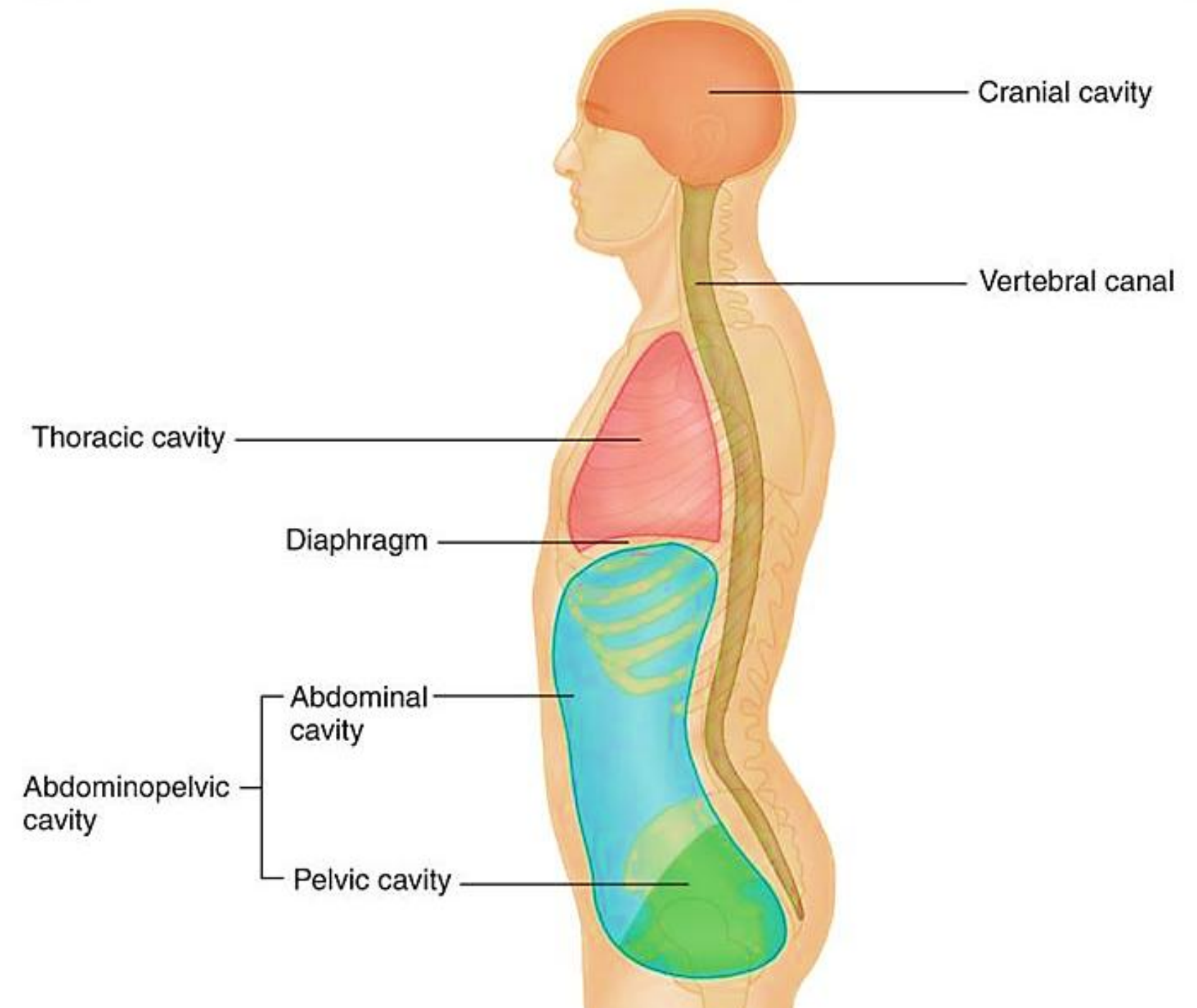


Body Cavities



Body cavities

- Spaces within the body which contain vital organs.



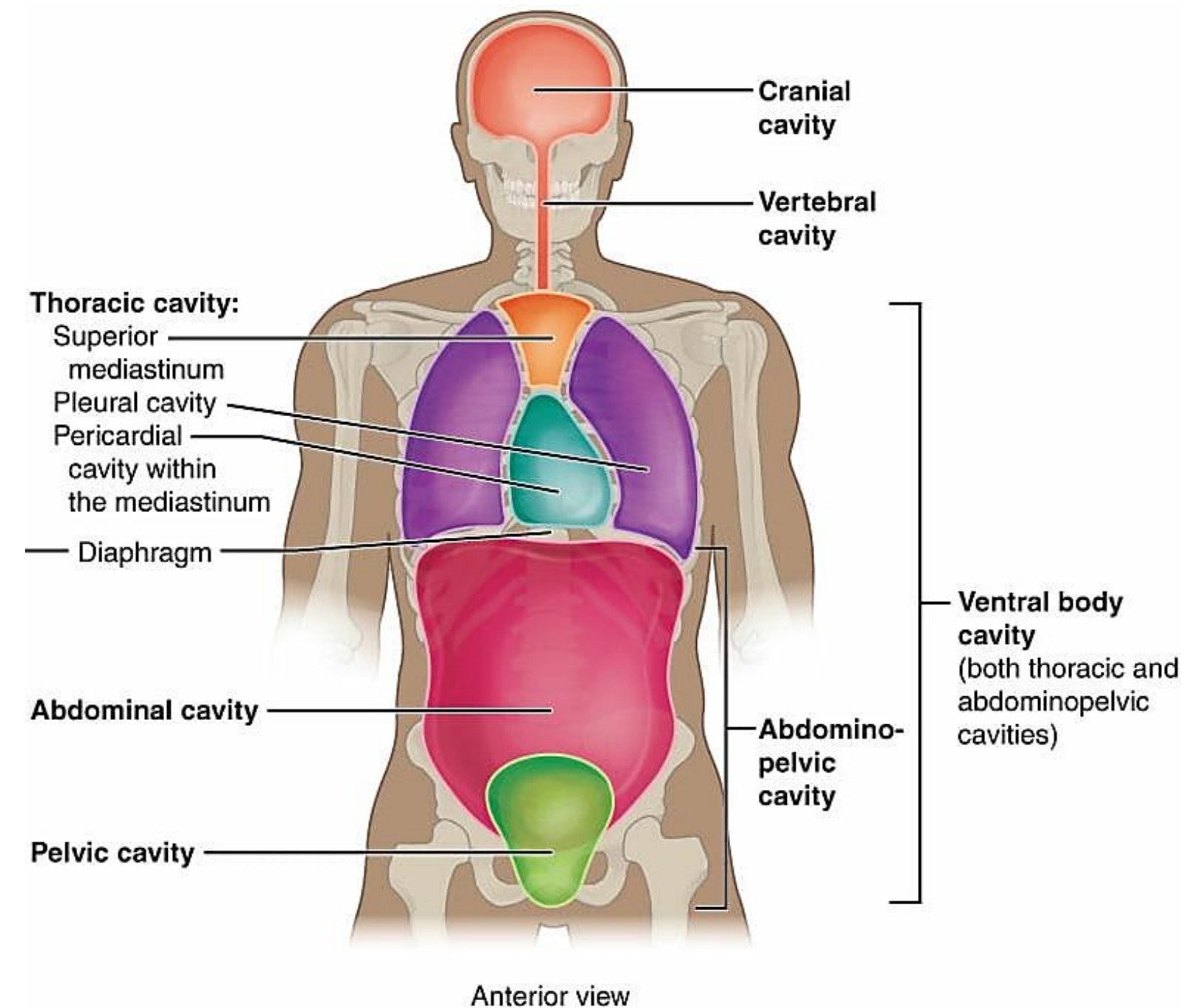
Body cavities

- I. Major cavities:** are divided into:
 - A. Dorsal cavities:**
include **cranial** and **vertebral cavities** which are continuous with each other at the base of skull through **foramen magnum**.
 - B. Ventral cavities:**
include **thoracic** and **abdominopelvic** cavities separated by the **diaphragm**.

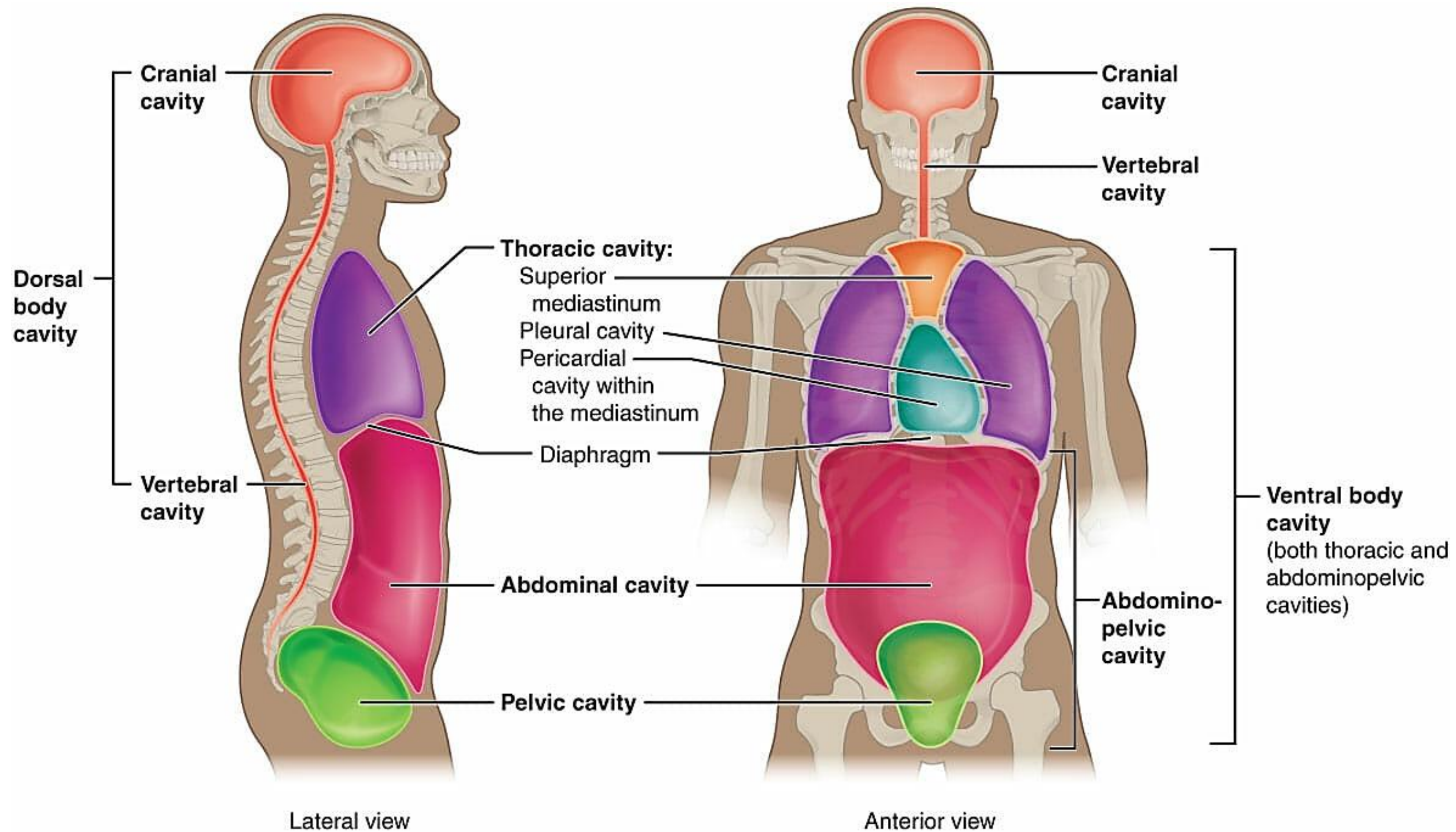
Body cavities

- **Thoracic cavity:**
Subdivided into two **lateral cavities** and a **median central cavity** called **mediastinum**.

Each lateral thoracic cavity contains the **lung** and its covering **pleura**.

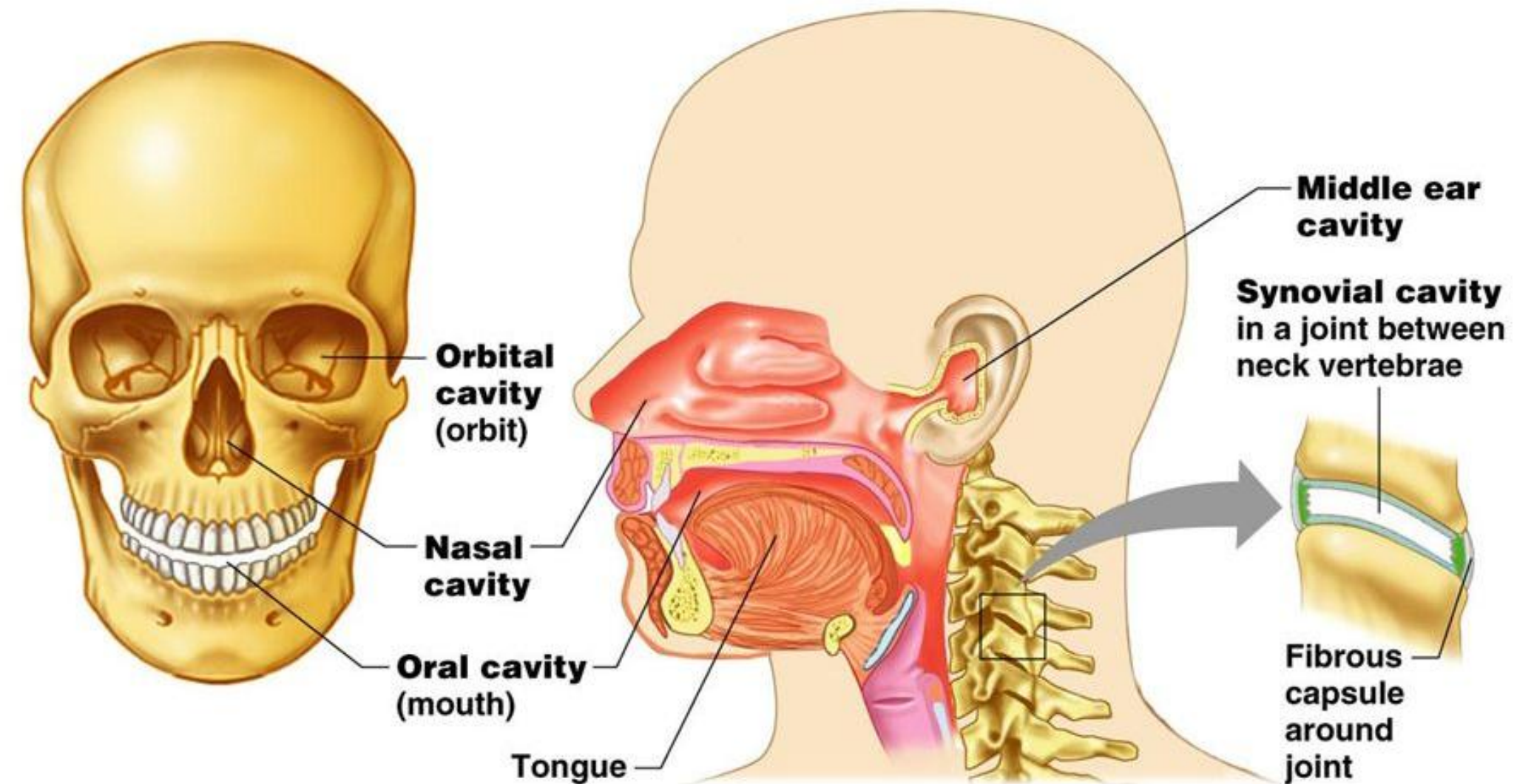


Body cavities



Body cavities

II. Minor cavities:
include nasal, orbital,
ear, oral and synovial
cavities.



Integumentary system

Skin

SKIN

- The skin, the body's largest organ, consists of the **epidermis**, a superficial cellular layer, and the **dermis**, a deep connective tissue layer.



SKIN

Functions of the skin

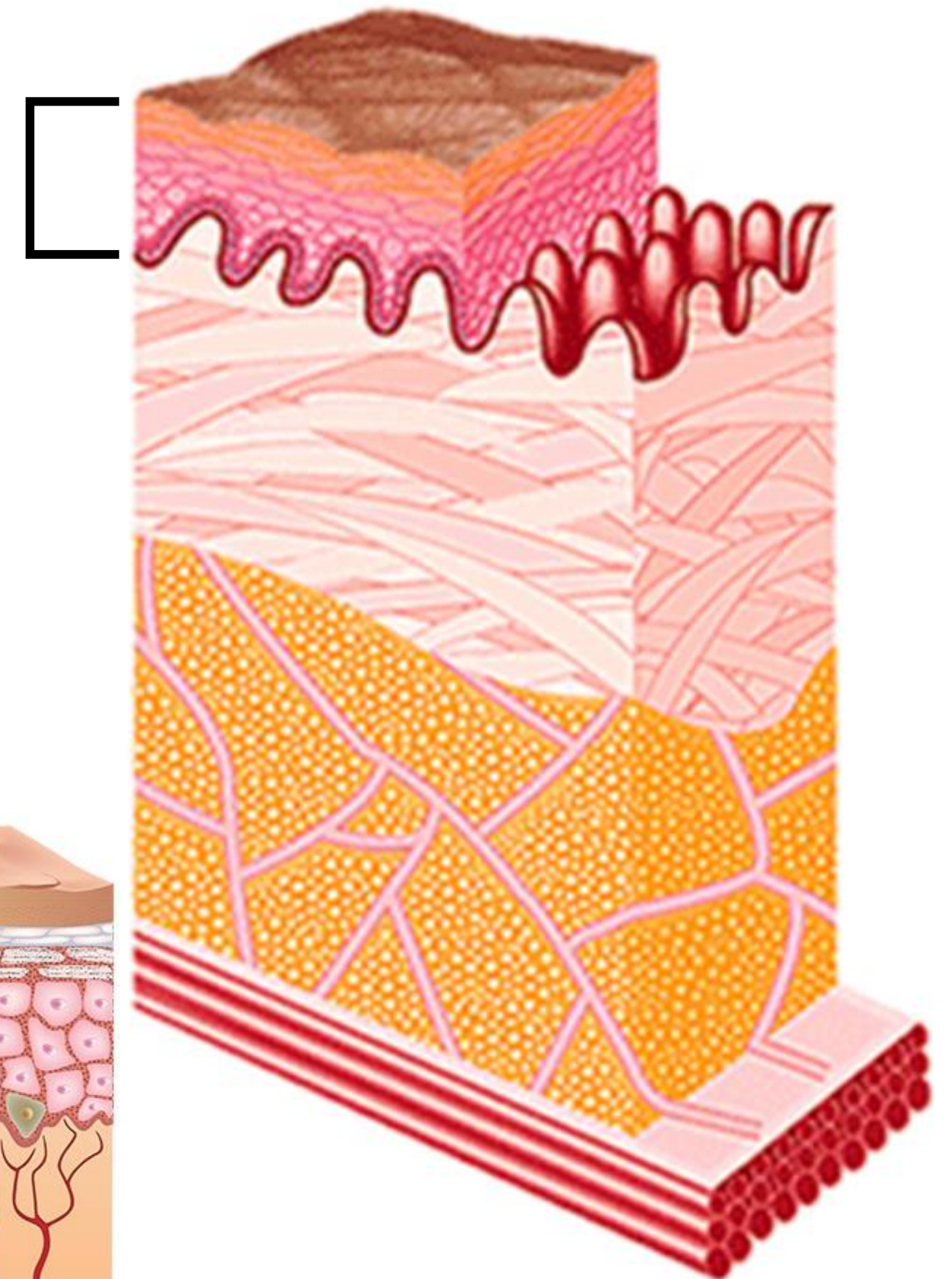
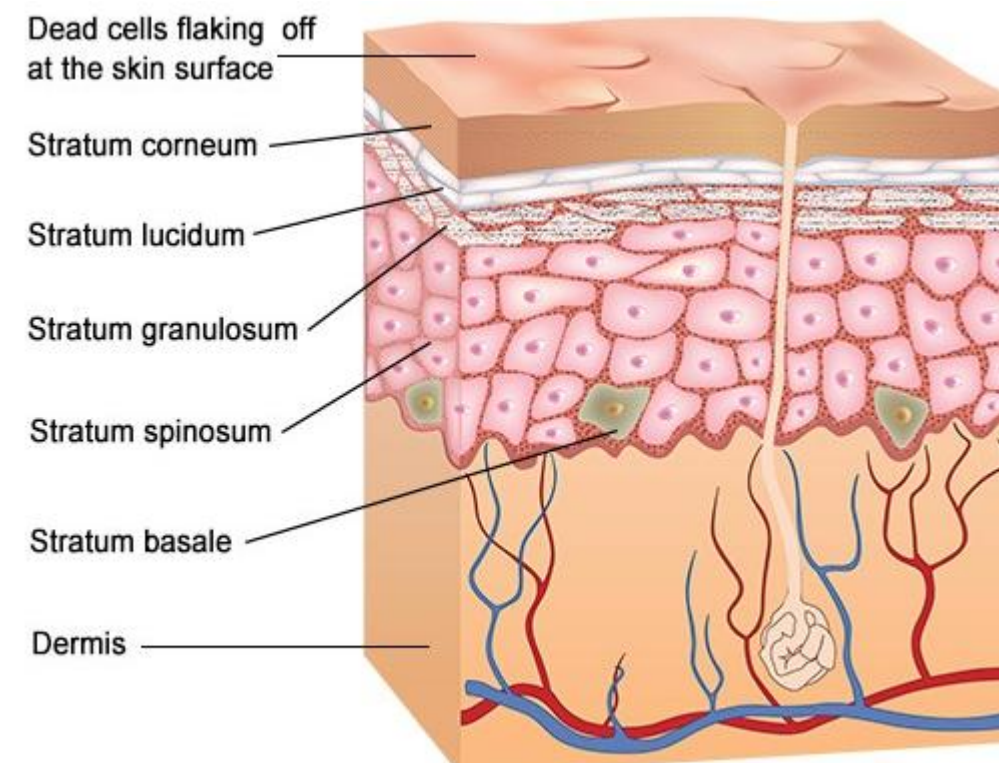
- Protection
- Heat regulation
- Sensation
- Synthesis and storage of vitamin D.



SKIN

■ EPIDERMIS

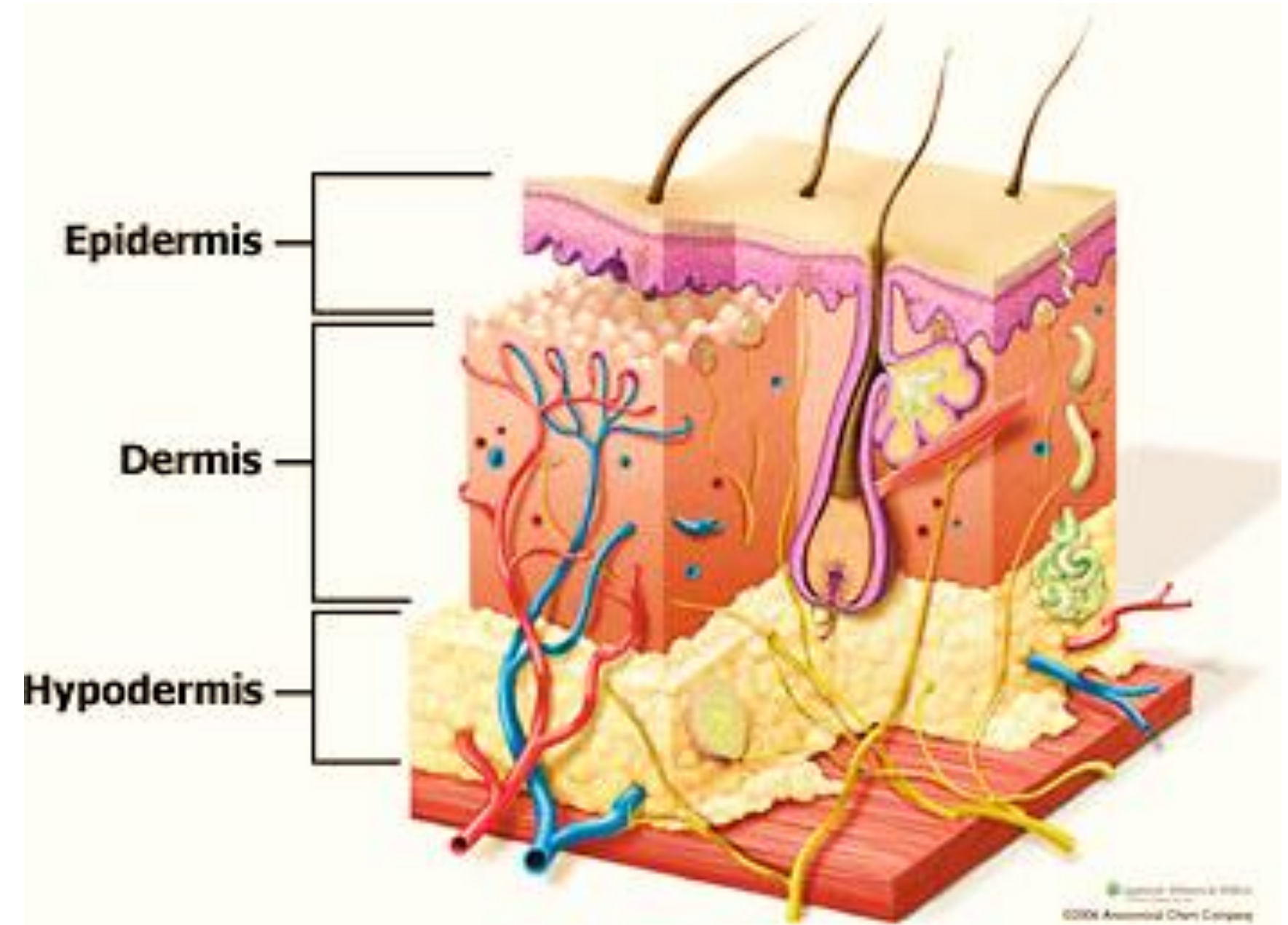
- Stratified squamous epithelium.
- A superficial keratin (dead tissue).
- A deeper regenerative cellular layer.
- Its non-vascular layer.



SKIN

■ DERMIS

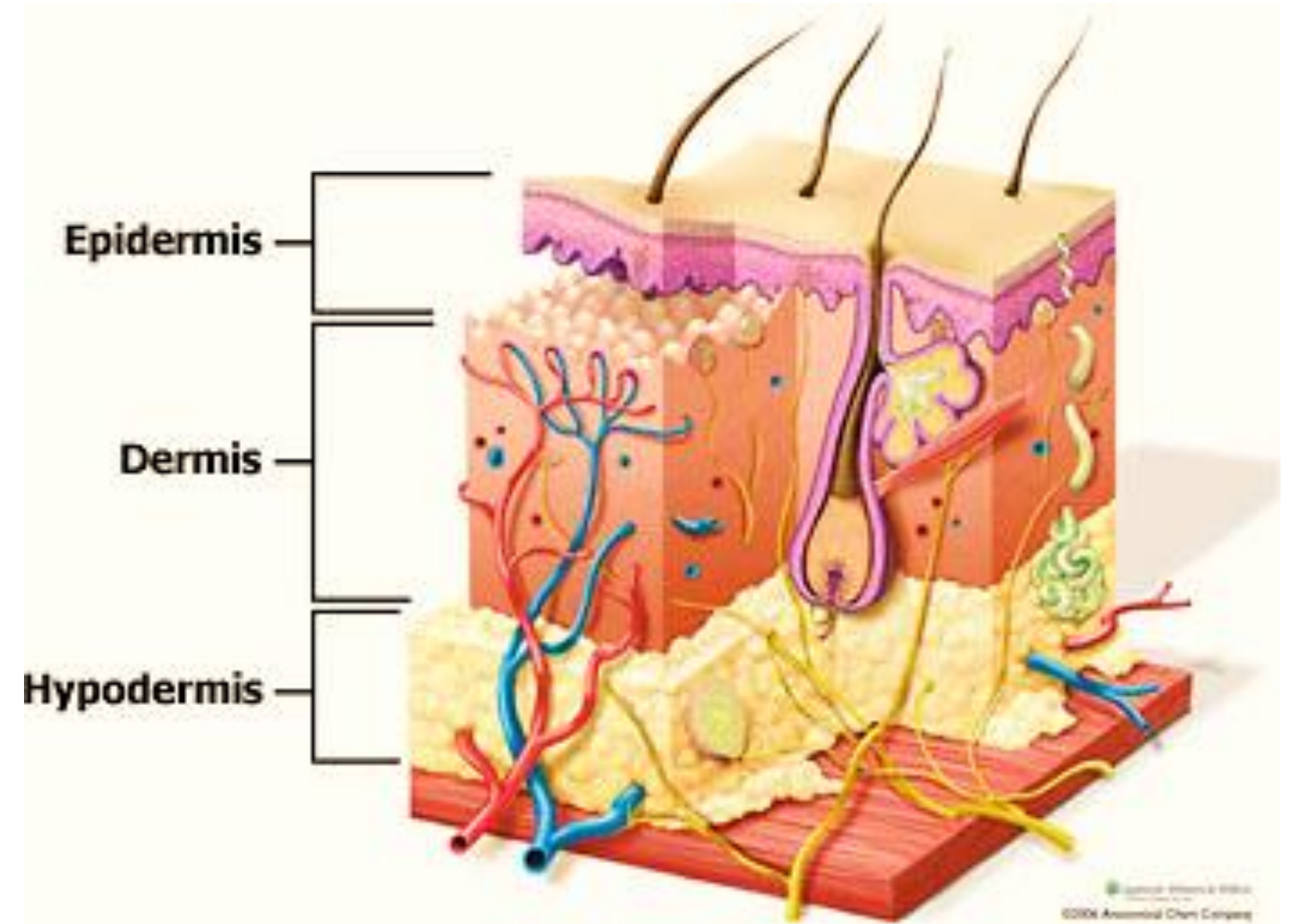
- Deep to epidermis.
- Consists of **collagen & elastic fibers**.
- **Contains:**
 1. Hair follicles,
 2. Arrector pili muscle,
 3. Sebaceous glands.



SKIN

■ Subcutaneous tissue

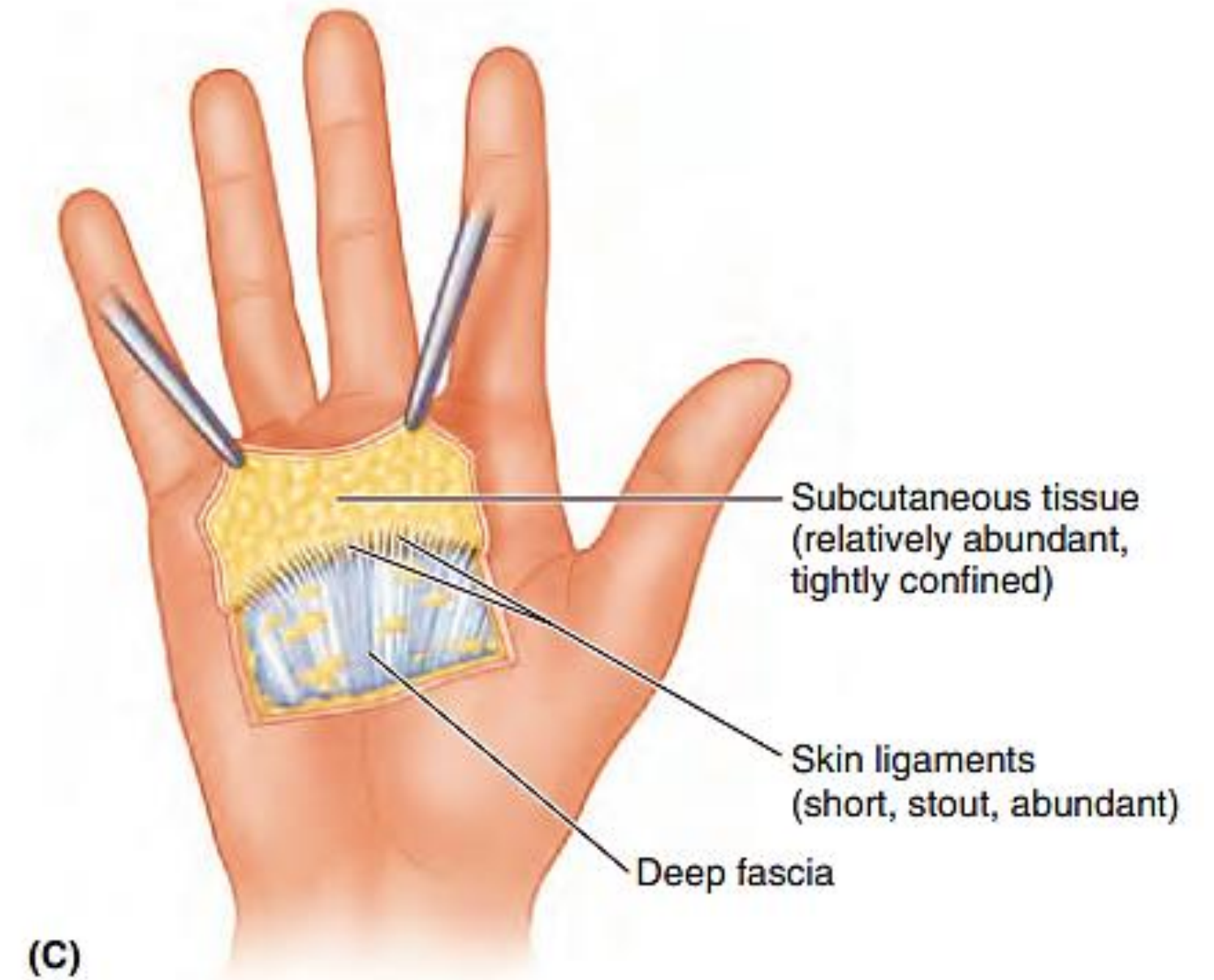
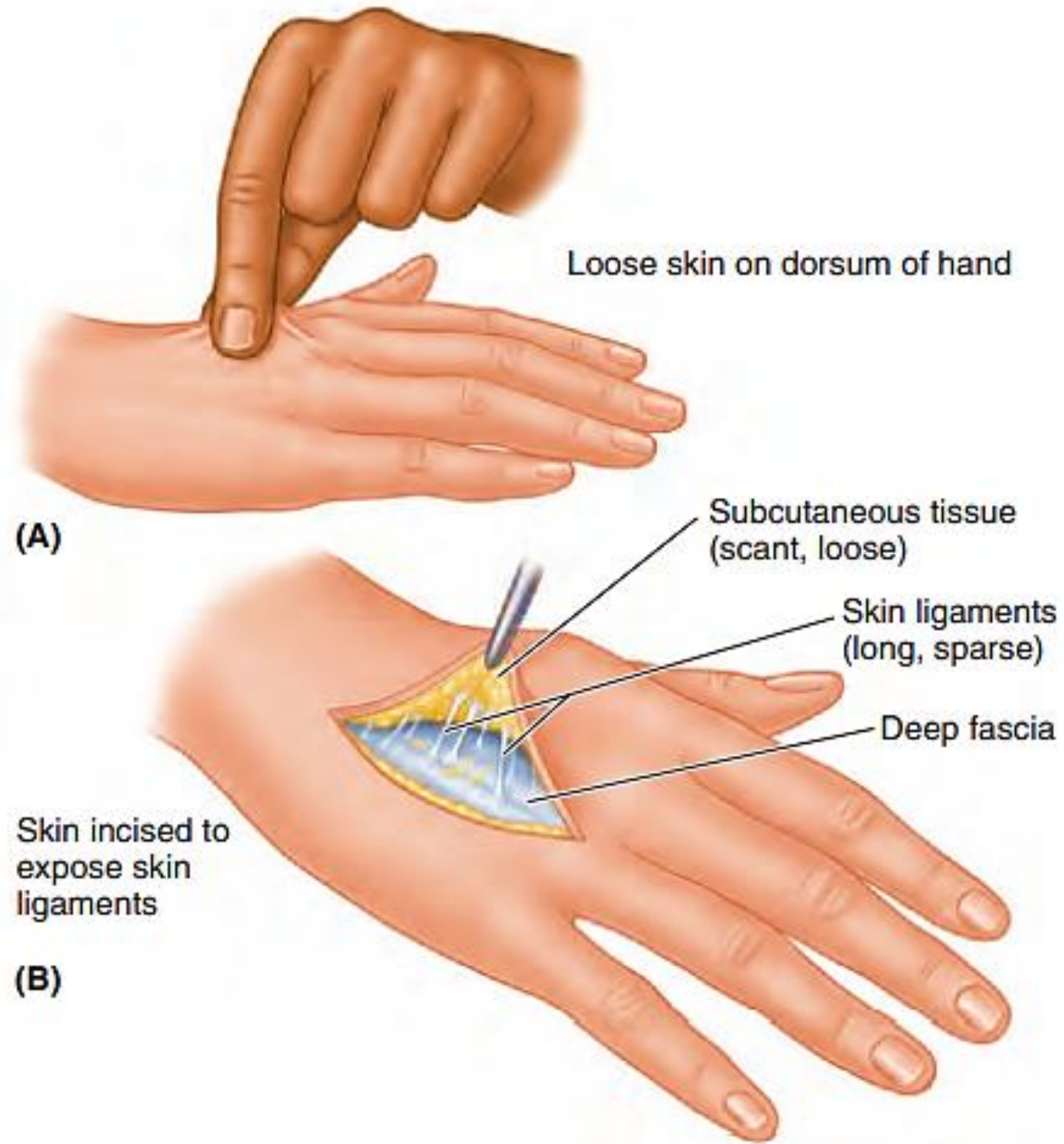
- also known as **Superficial fascia**
- **Composed of :**
 1. Loose connective tissue
 2. Stored fat
 3. Sweat glands
 4. Skin ligaments
 5. Superficial :
 6. Blood vessels
 7. Lymphatic vessels
 8. Cutaneous Nerves



SKIN

- **Skin ligaments (of Subcutaneous tissue)**
 - In the **back of the neck, scalp, palms of hands, and soles of feet**, the subcutaneous tissue contain numerous **bundles of collagen holding the dermis of the skin firmly to deeper structures.**
 - In the **eyelids, auricle, penis and scrotum and clitoris**, its **devoid of fat.**

SKIN



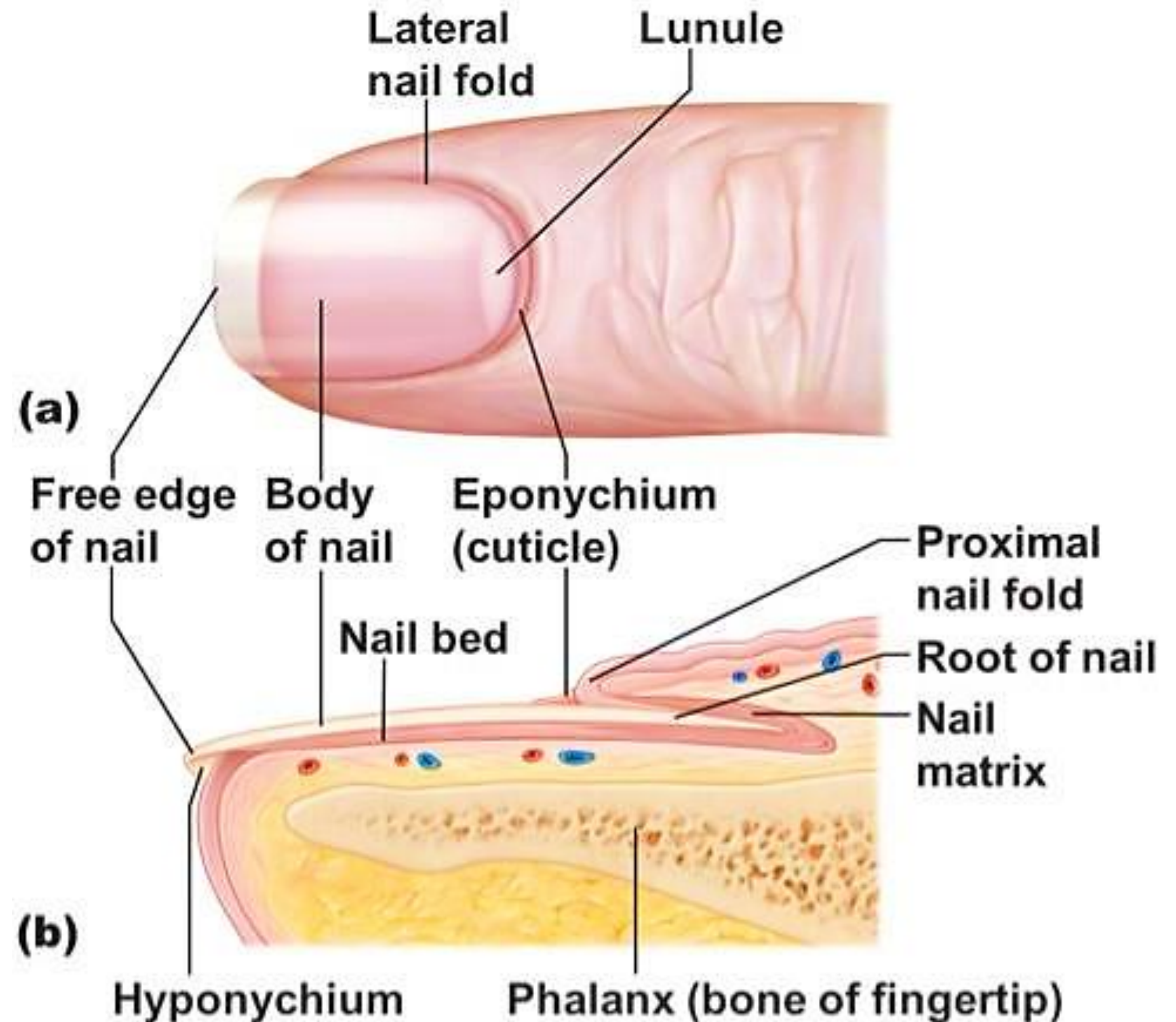
SKIN

- **SKIN APPENDEGES**

1. Nails
2. Hair follicles
3. Sebaceous glands
4. Sweat glands (including mammary glands)

SKIN APPENDEGES

1. **NAIL:** keratinized plates at dorsum of fingers and toes.



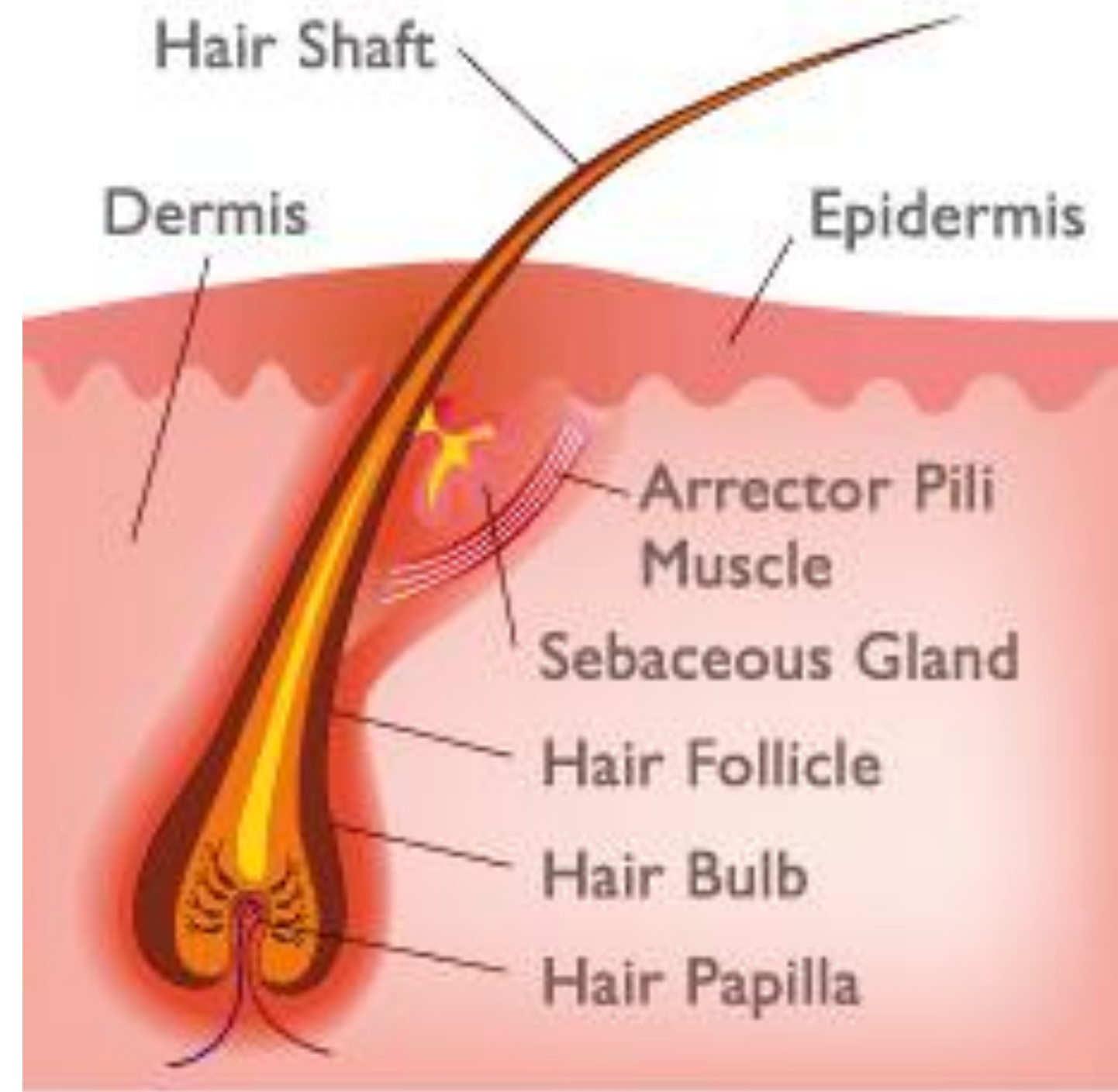
SKIN APPENDEGES

2. HAIR FOLLICLES:

- not present in lips, palms, side of fingers, penis and clitoris.

3. SEBACEOUS GLAND:

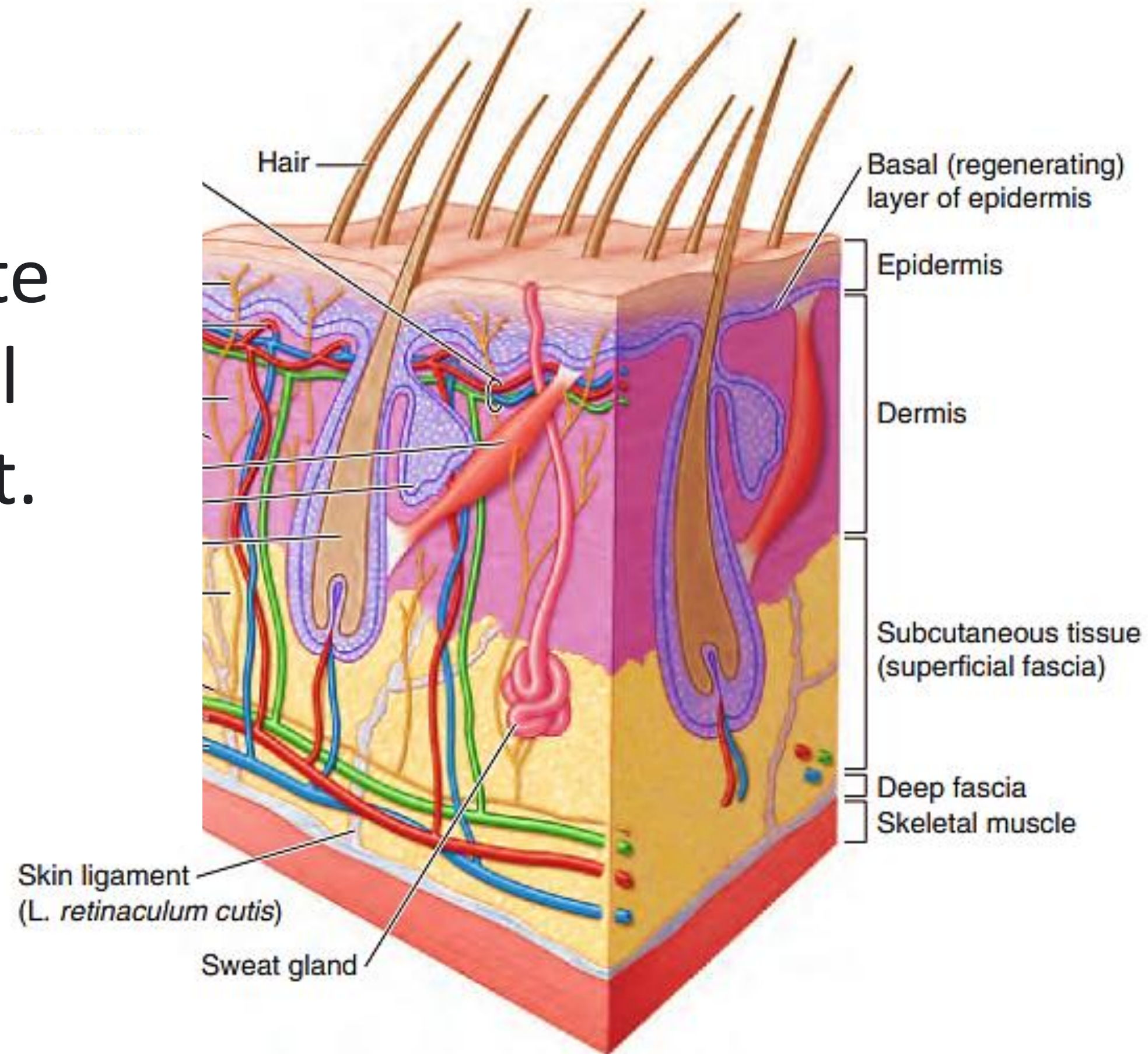
- opens into a hair follicle to secrete an oily or waxy matter, called sebum, which lubricates the hair and skin.



SKIN APPENDEGES

4. SWEAT GLANDS

- that produce and secrete **sweat** onto an epithelial surface by way of a duct.



SKIN

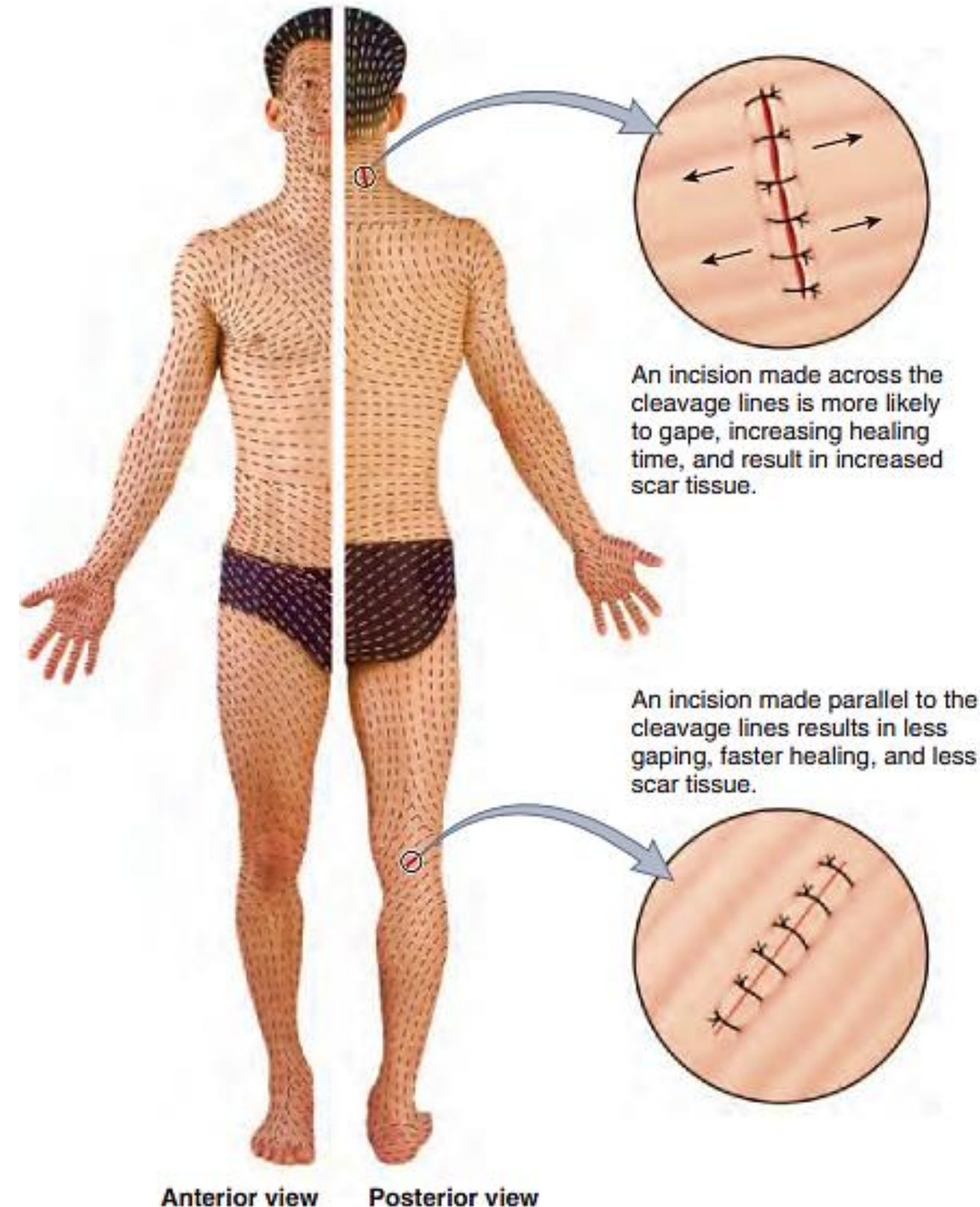
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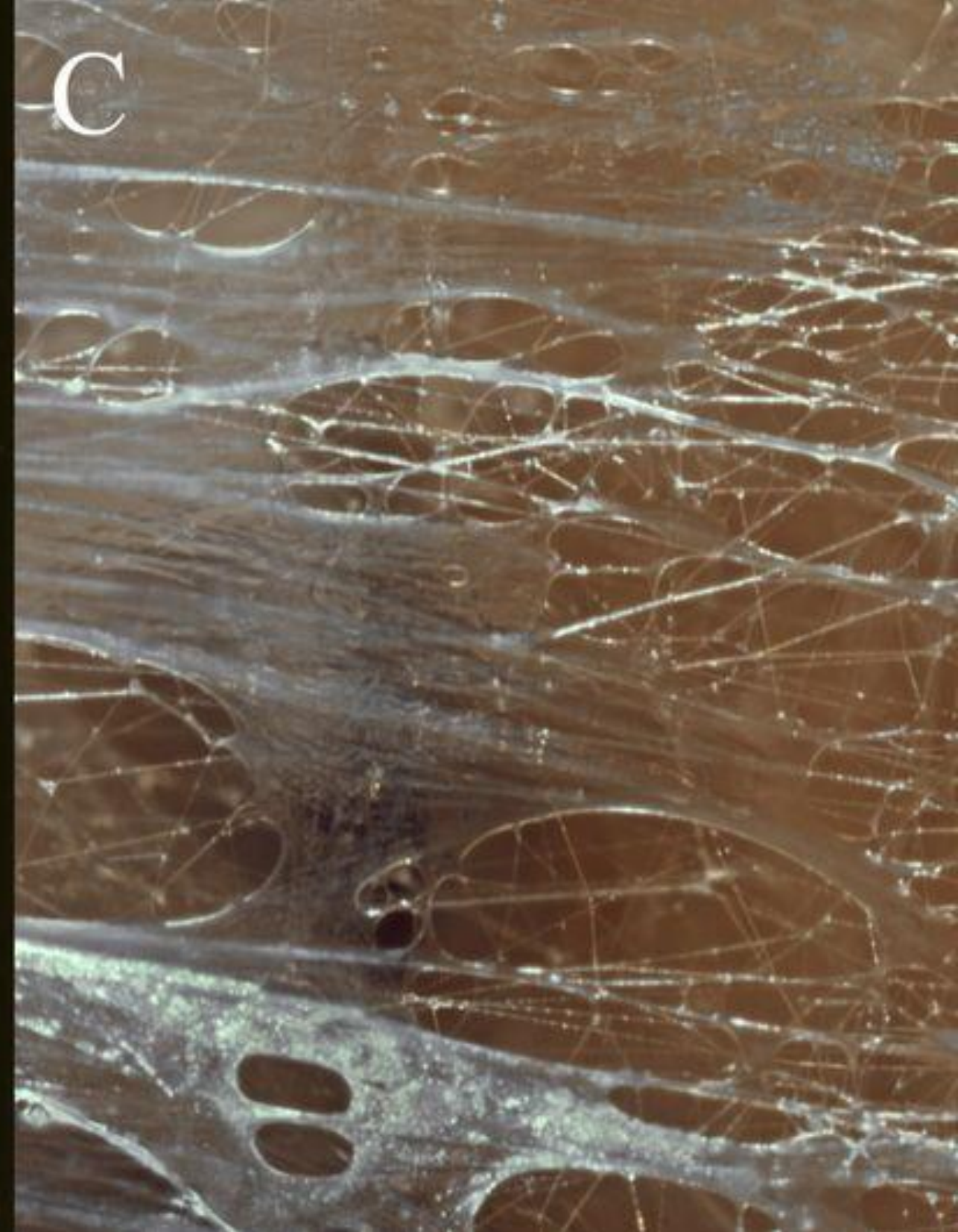
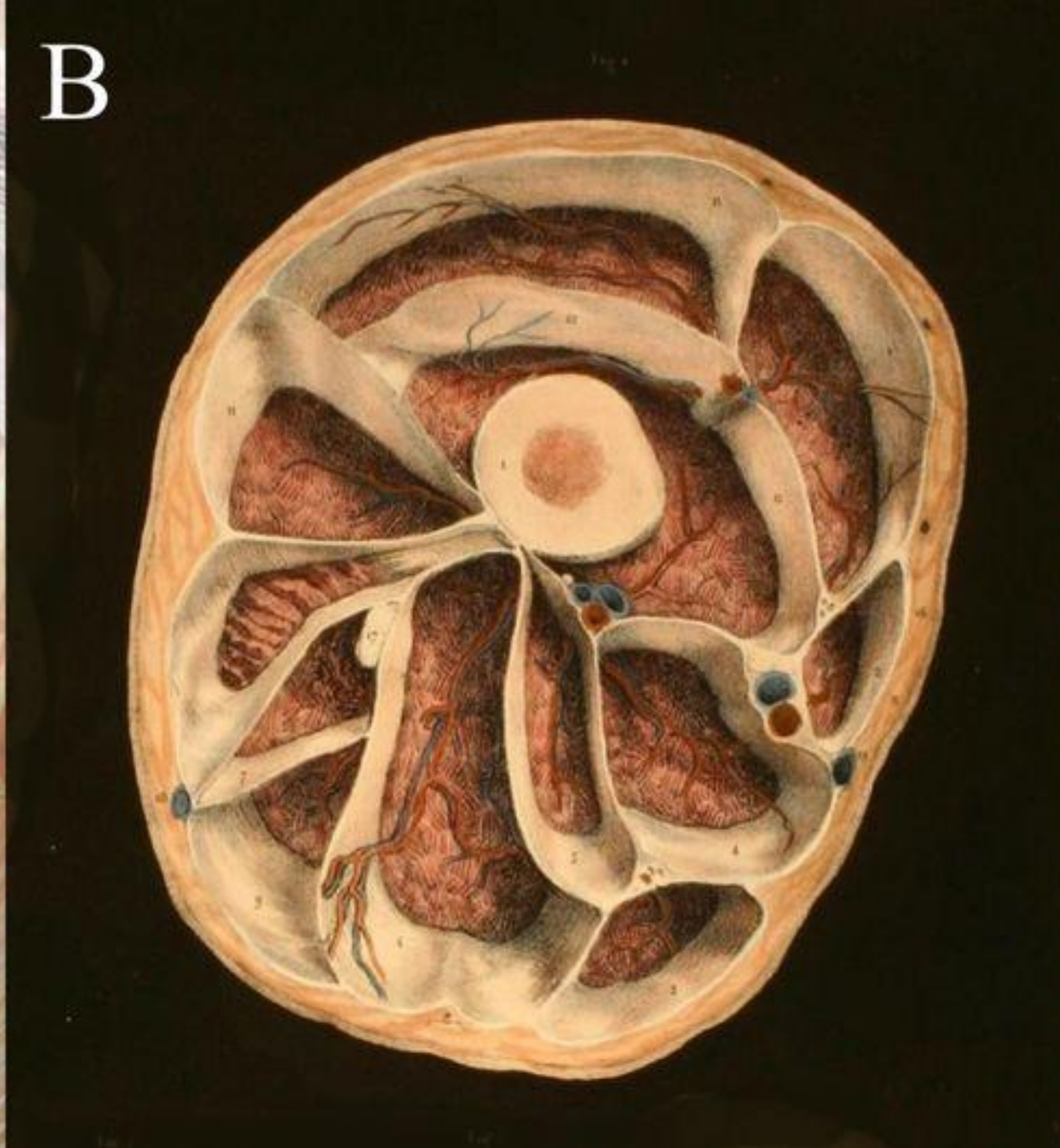
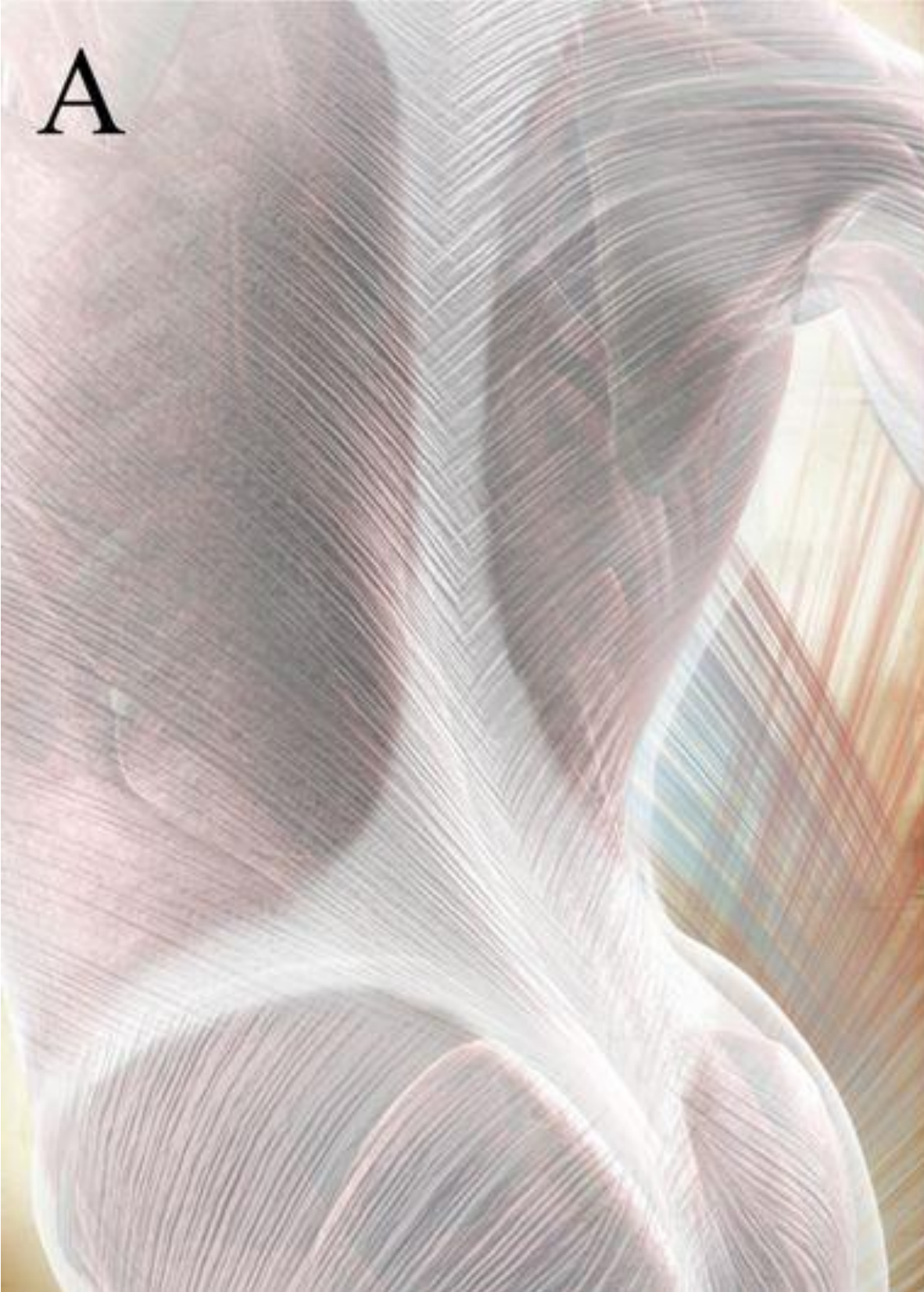
- The **skin thickness** varies in different parts of the body and is **thinner in women than in men**.
- The epidermis is **thicker in the palms and sole** of the feet.
- The dermis is **thicker on the posterior aspect of the body than anteriorly**.

SKIN

■ LANGER'S LINES

- The direction of the rows of the collagen is known as **lines of cleavage (Langer's lines)** and they tend to run **longitudinally in the limbs** and **circumferentially in the neck and trunk**.
- Incisions parallel to these lines tend to **heal with lesser scar formation**.





FASCIAE

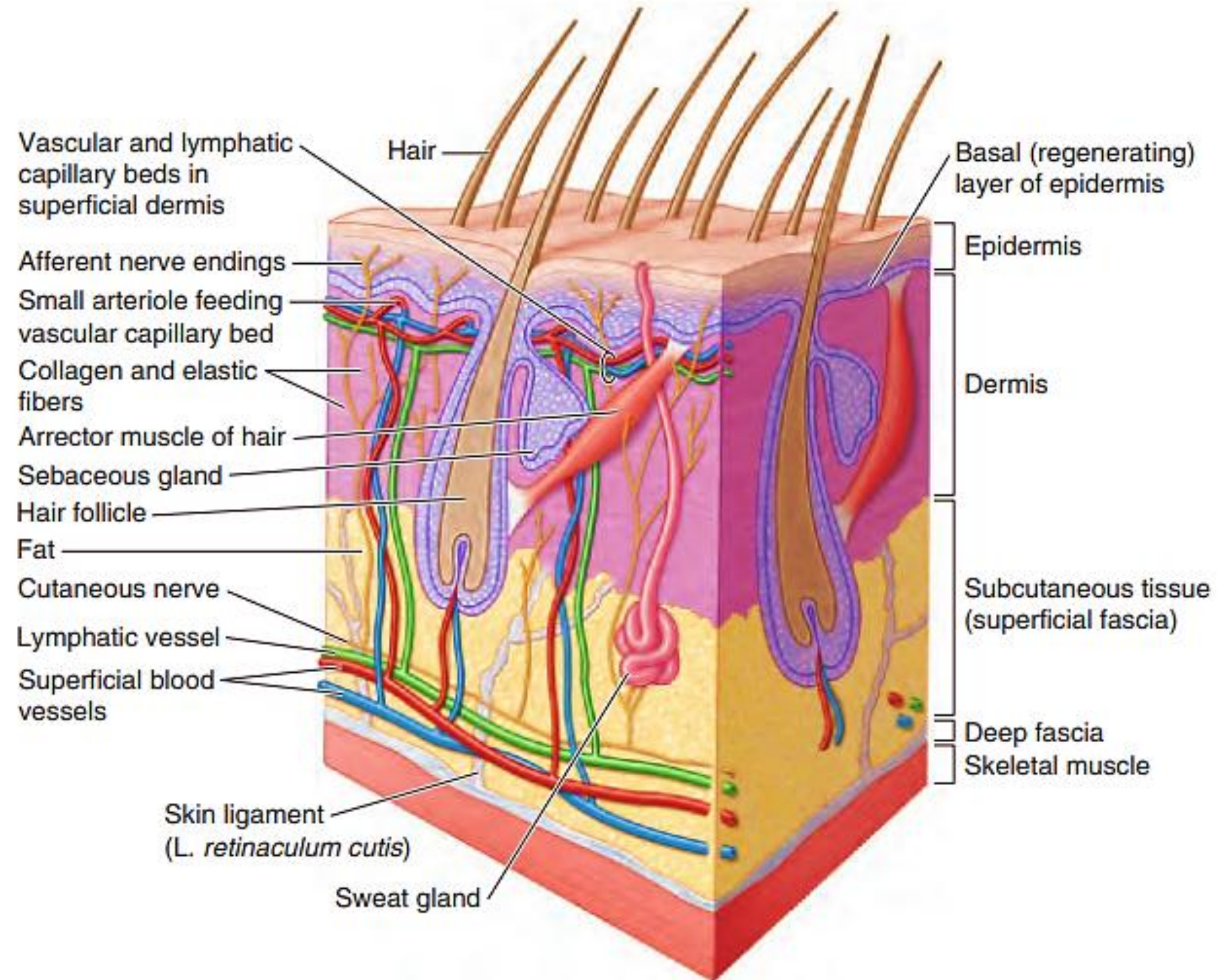
FASCIA

- **Constitute the wrapping, packing, and insulating materials** of the deep structures of the body.
 1. Superficial fascia (its subcutaneous tissue).
 2. Deep fascia.
 3. Intermuscular septa.
 4. Retinacula.
 5. Bursae.

FASCIA

2. DEEP FASCIAE:

- A layer of **dense connective tissue** which can surround individual muscles.



FASCIA

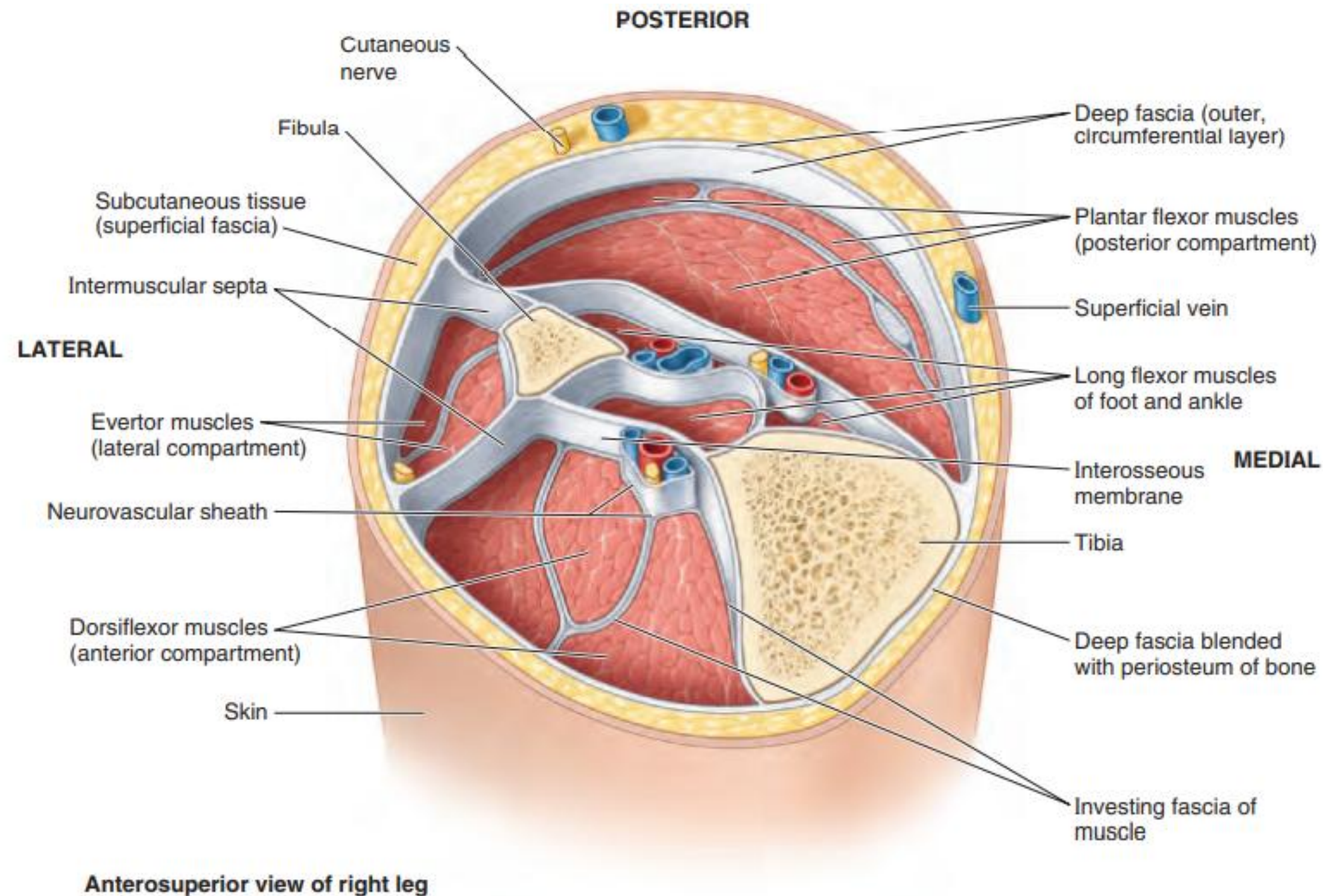
2. DEEP FASCIAE:

- And also **surround groups of muscles** to separate into fascial compartments.
- **Devoid of fats.**
- This fibrous connective tissue **interpenetrates and surrounds** the muscles, bones, nerves and blood vessels of the body.
- These extensions known as **Investing fascia**.

FASCIA

3. Intermuscular septa.

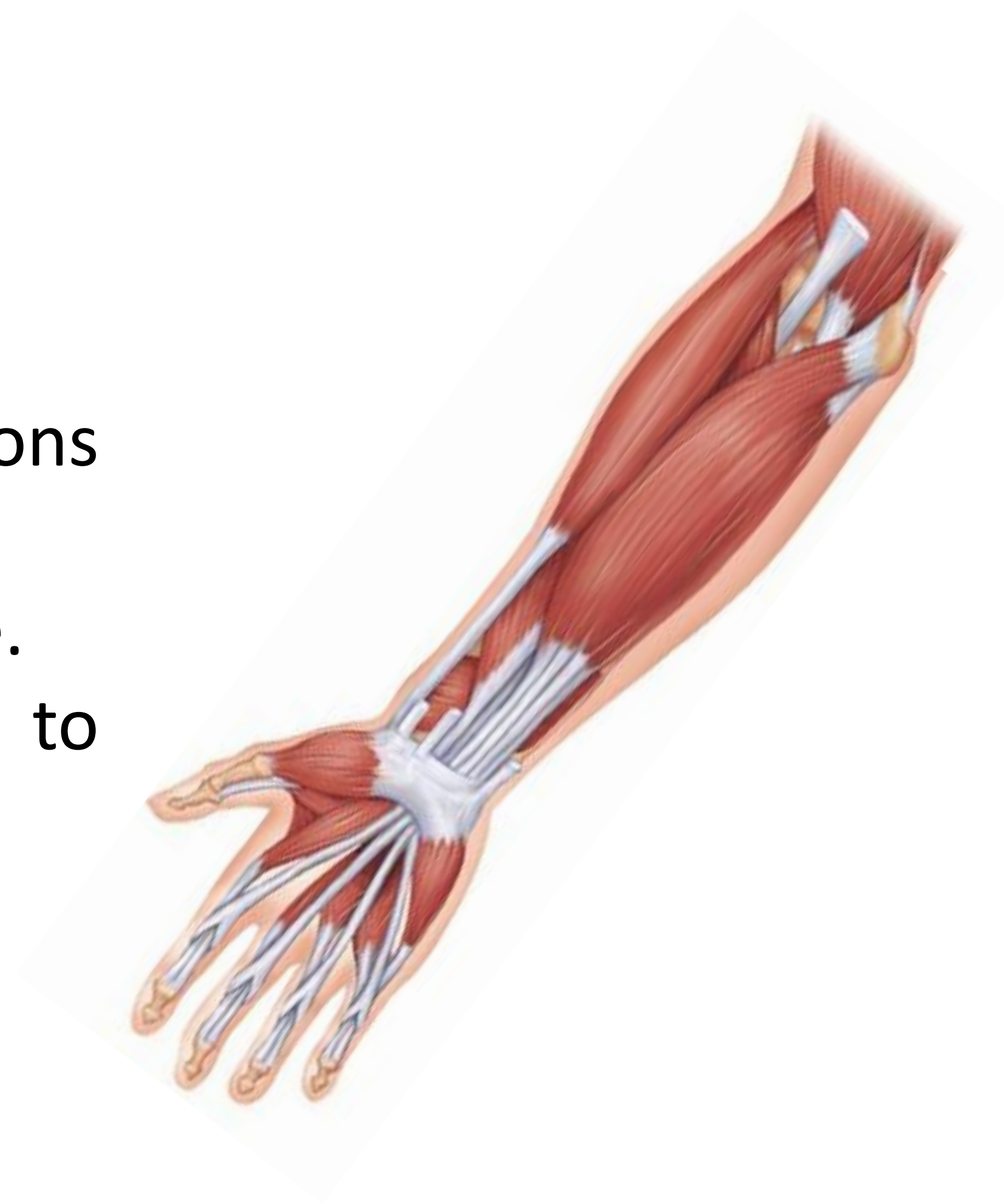
- In the limbs, groups of **muscles with similar functions** sharing the same nerve supply are located in fascial compartments, separated by **thick sheets of deep fascia, called intermuscular septa**, that extend centrally from the surrounding fascia to attach to bones.



FASCIA

4. RETINACULA

- is a band around tendons that holds them in place.
- It is not part of any muscle.
- Its function is mostly to stabilize a tendon.



Thank You

